2026 Fall

# CS270: Digital Image Processing

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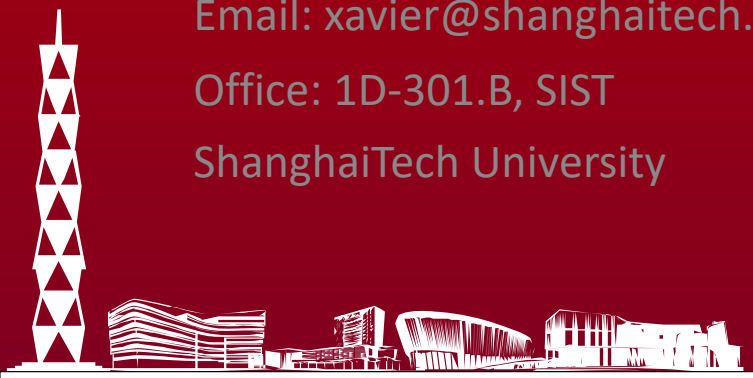
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# Lecture 3
# Intensity transformation

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2026 Fall – Sept 23th

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# Intensity Transformation: Outline



- Histogram （直方图）
  - Definition
  - Properties
- Intensity Transformation（灰度变换）
  - Linear transform
  - Non-linear transform
- Histogram Processing（直方图处理）
  - Histogram Equalization
  - Histogram Matching



# Definition



## ❑ Histograms

$$h(r_k) = n_k$$

Where $r_k$ : the $k^{th}$ intensity value in the level range of [0, L-1]

$n_k$ : the number of pixels in the image with intensity $r_k$

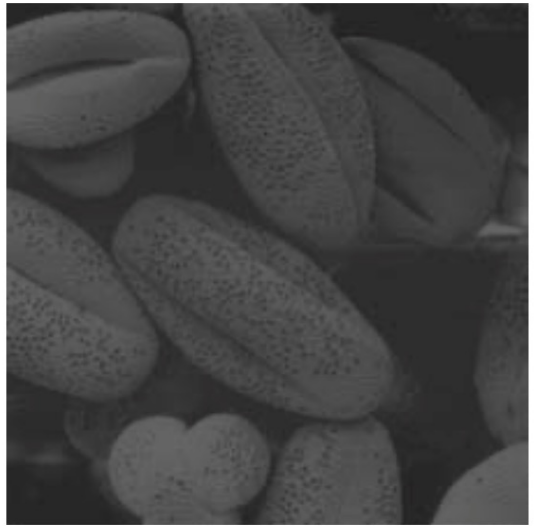
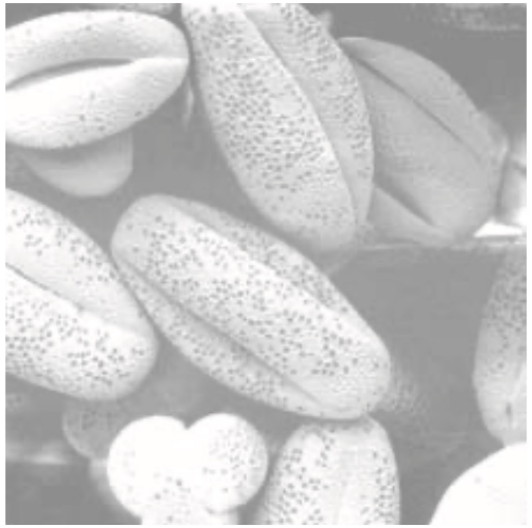
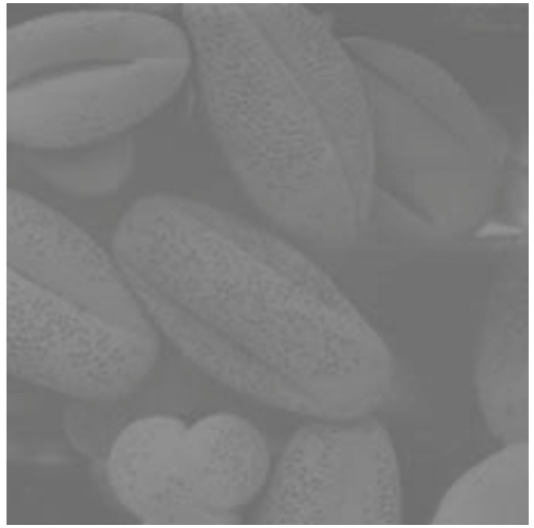
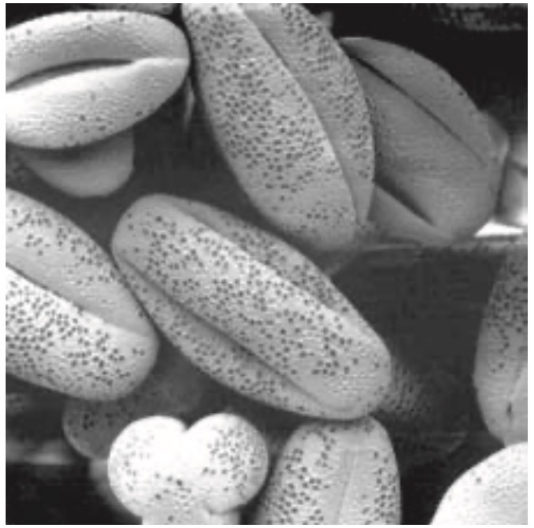
## ❑ Normalized Histogram（归一化直方图）

$$p(r_k) = \frac{h(r_k)}{M.N} = \frac{n_k}{M.N}$$

Where $p(r_k)$: the probability of occurrence of intensity $r_k$ in an image

$M, N$: the row and column dimensions of the image

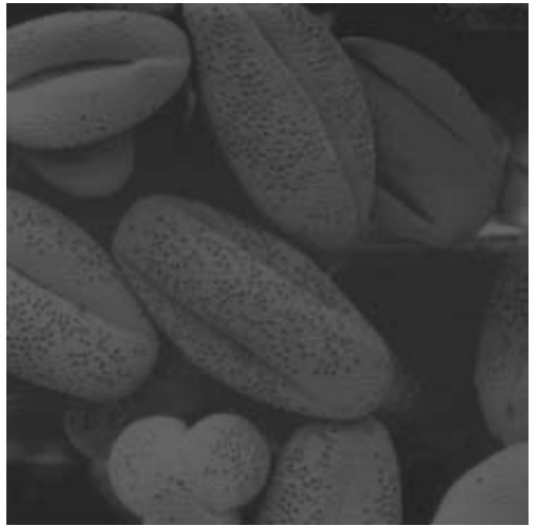
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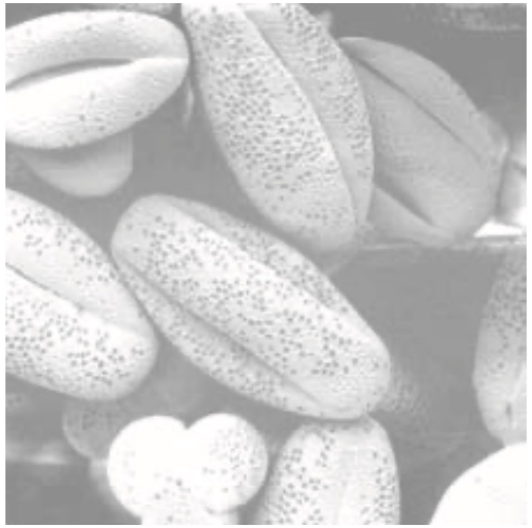





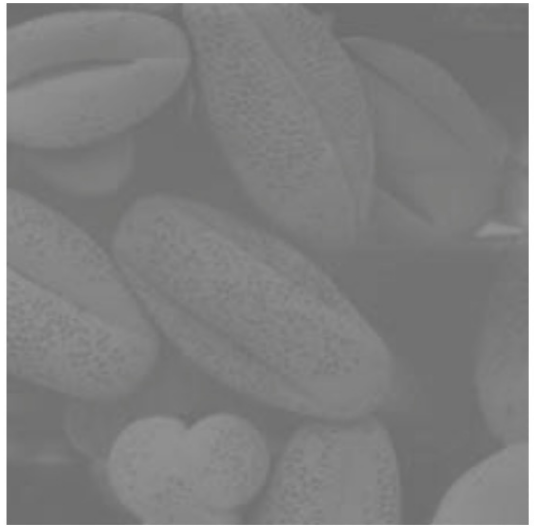
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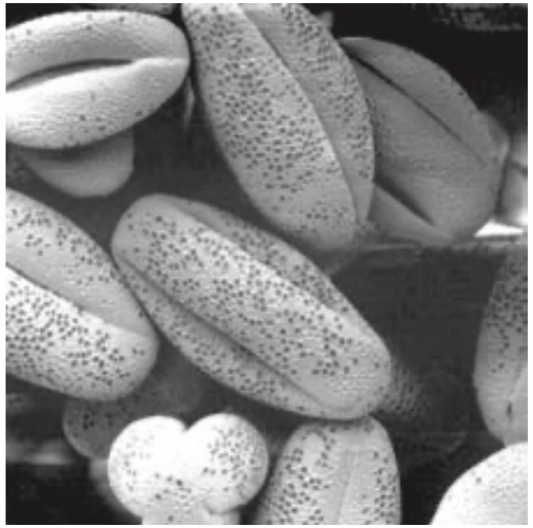
# Basic Image Type

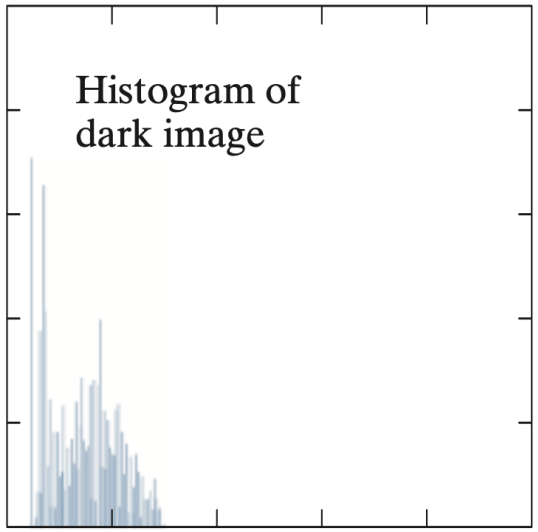


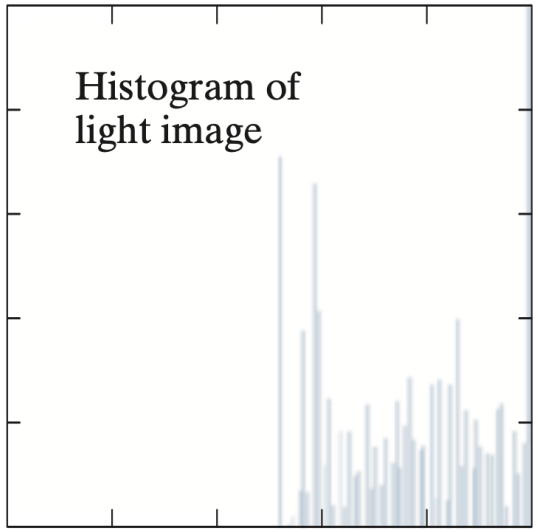


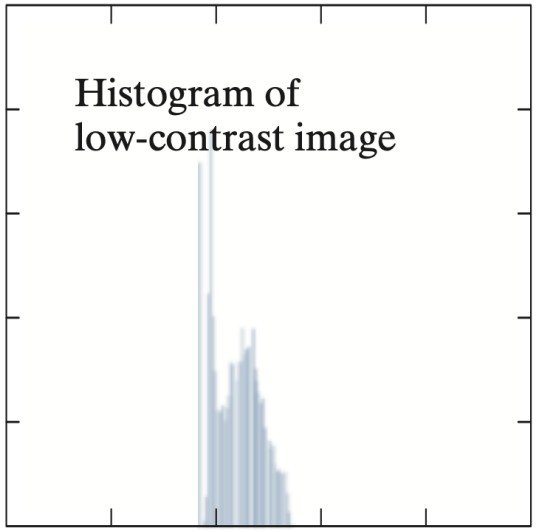


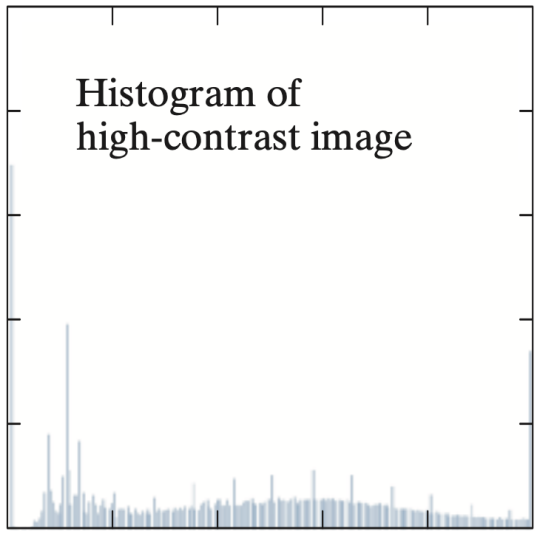






# Properties



- **The histogram of an image**
  - describes the number or probability of intensity, NO location (spatial) information
  - can be same as other images
  - $\sum_0^{L-1} n_k = M \cdot N$ or $\sum_0^1 p(r_k) = 1$
  - If Region C=AUB, A and B are disjoint, $H_C = H_A + H_B$

# Intensity Transformation



- **Simplest image processing techniques**

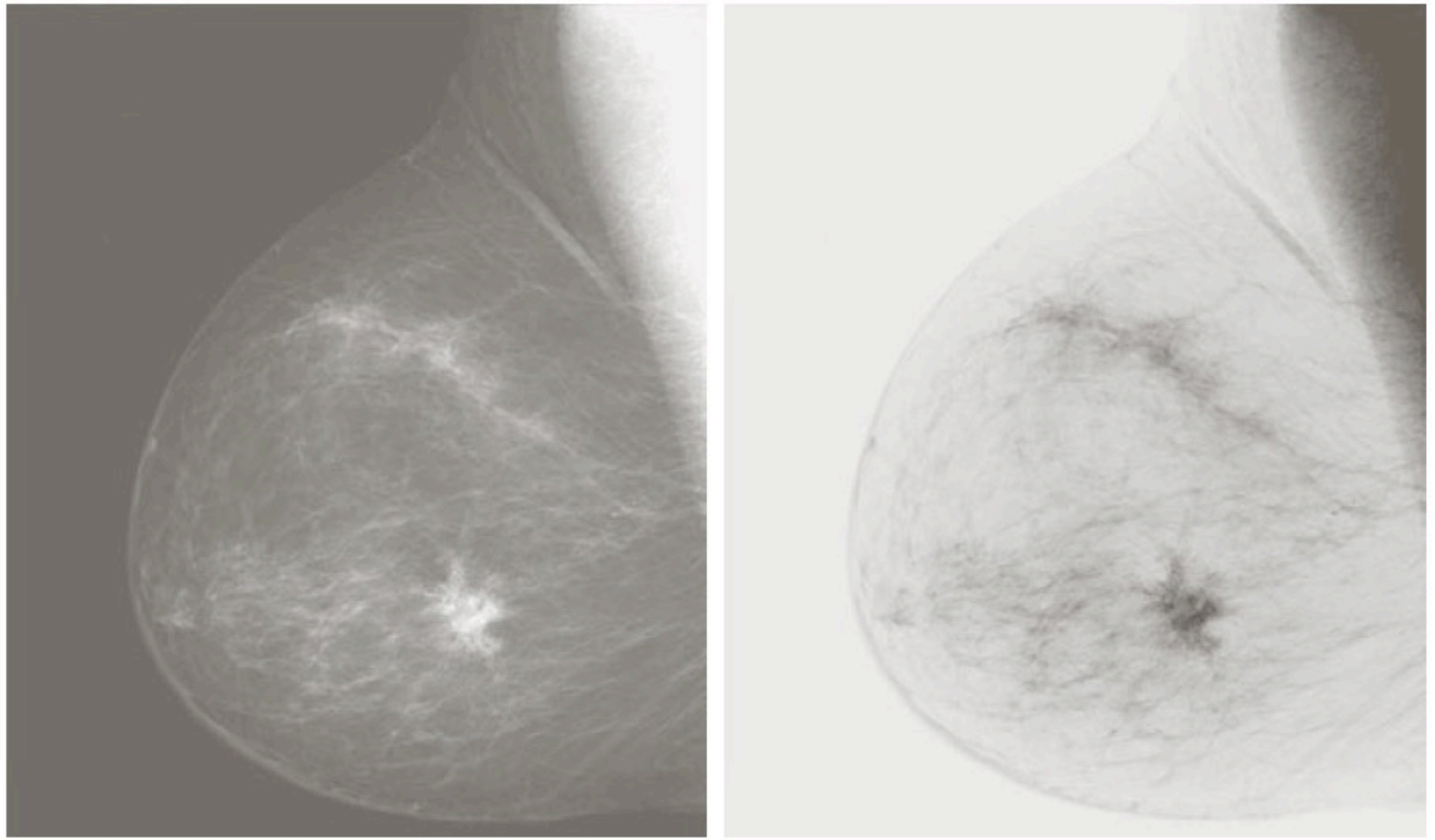
$$s = T(r)$$

- **Types of Intensity Transformation**
  - Image Negatives（图像反转）
  - Log Transformation（对数变换）
  - Power-law (gamma) Transformation（幂律/伽马变换）
  - Piecewise-Linear Transformation（分段线性变换）

# Image Negatives

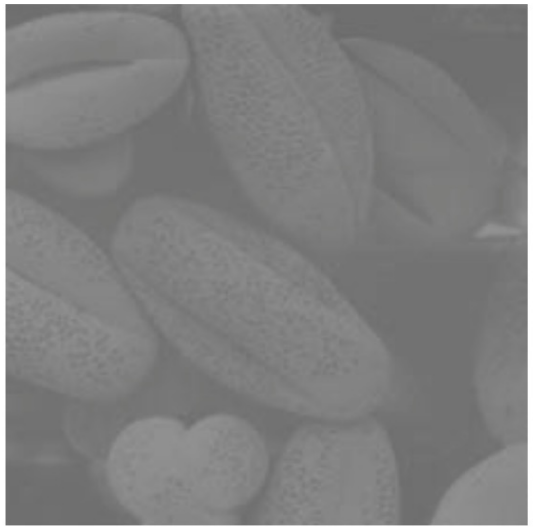


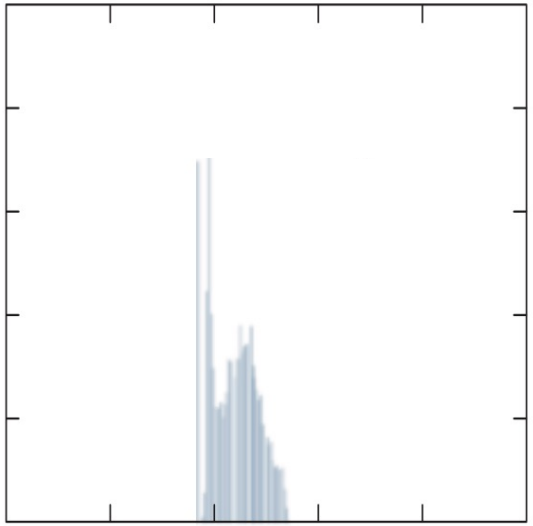
$$s = \mathrm{T}(r) = L - 1 - r$$

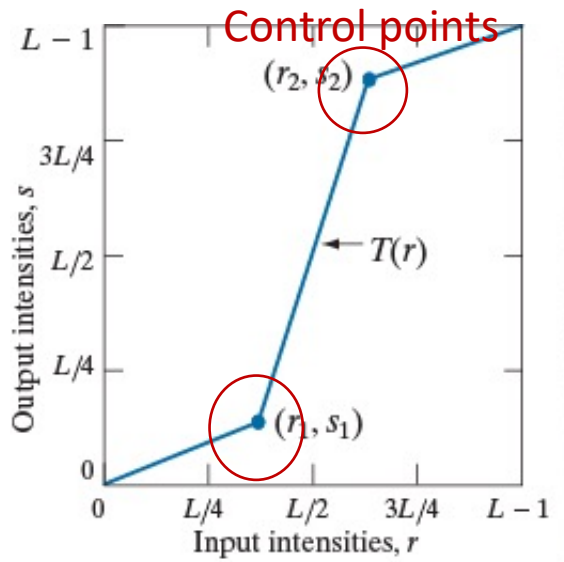


# Contrast Stretching



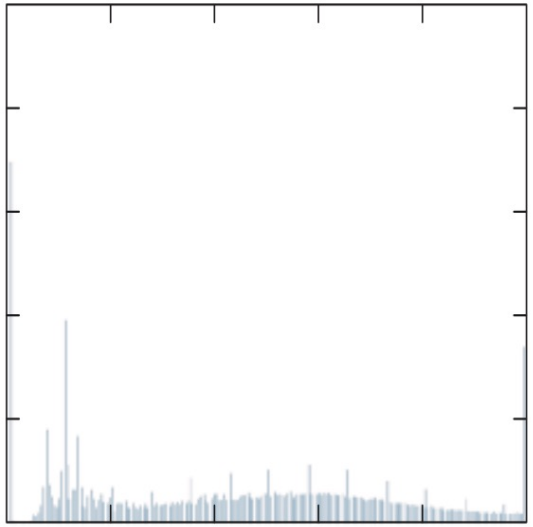







Usually: $r_1 \leq r_2$ and $s_1 \leq s_2$



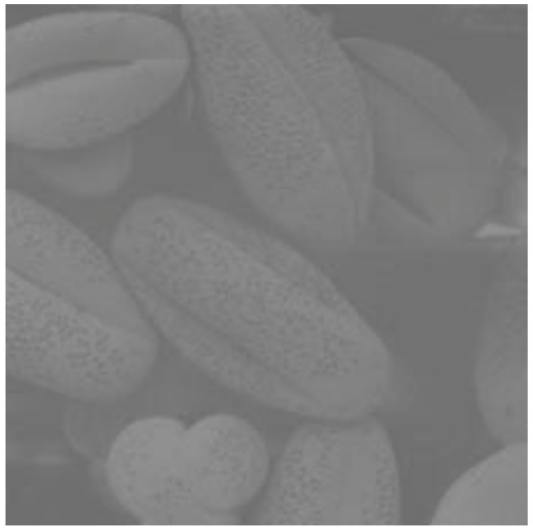


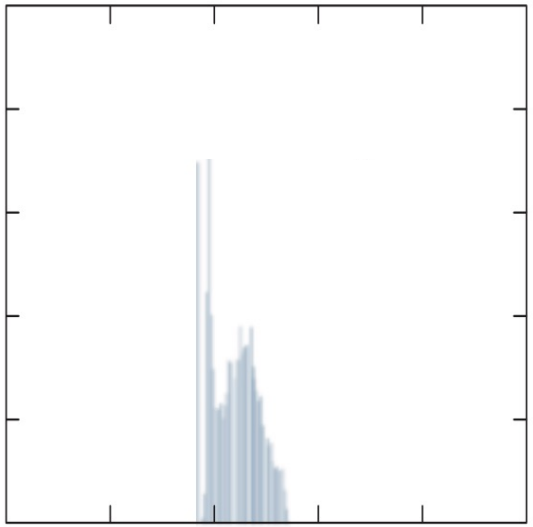
Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times.

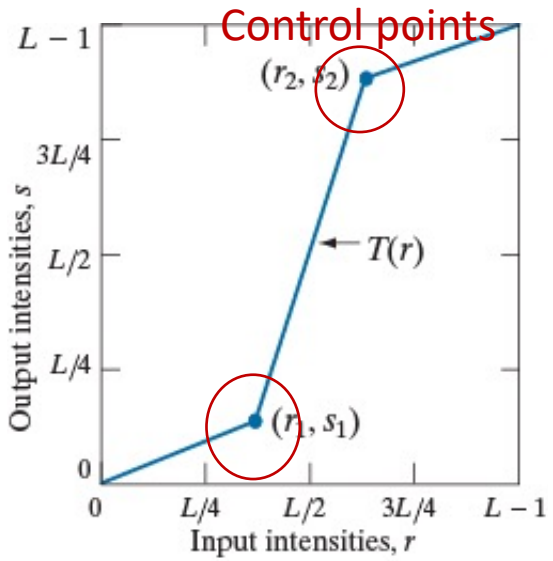
(Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

# Contrast Stretching



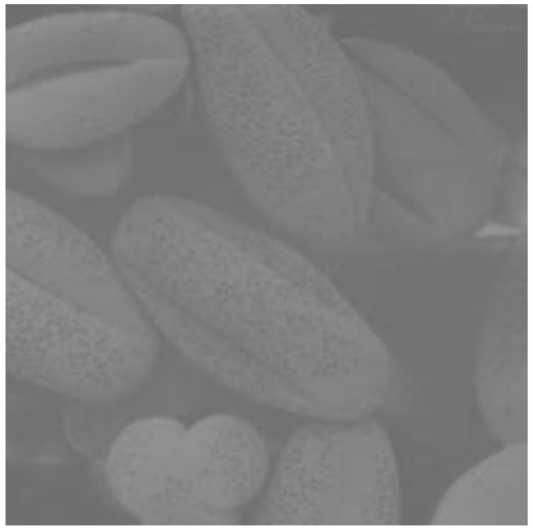


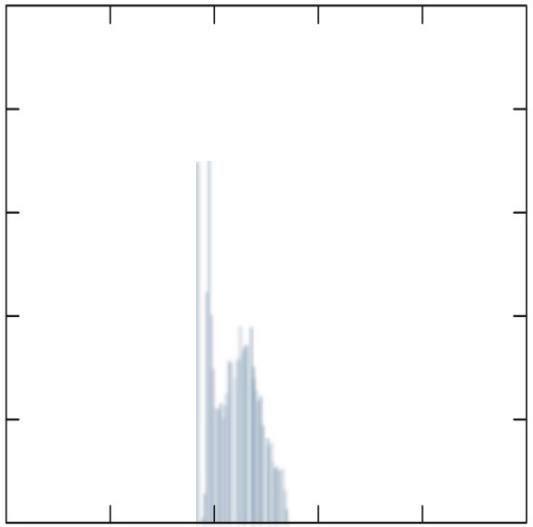


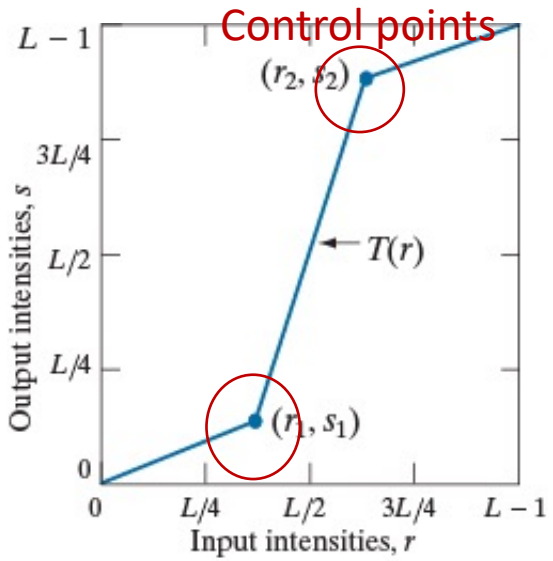



$s = \mathrm{T}(r) = ?$

# Contrast Stretching



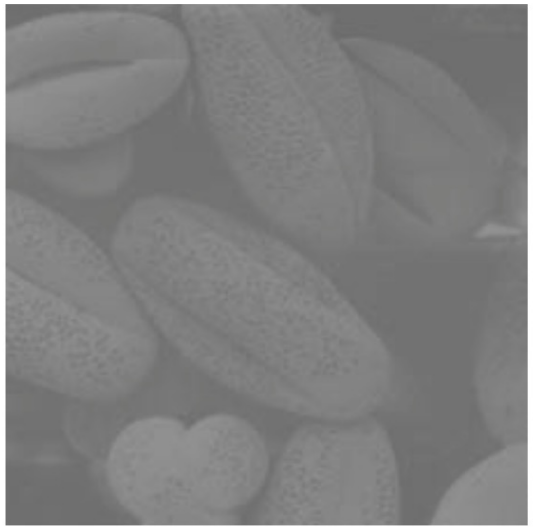


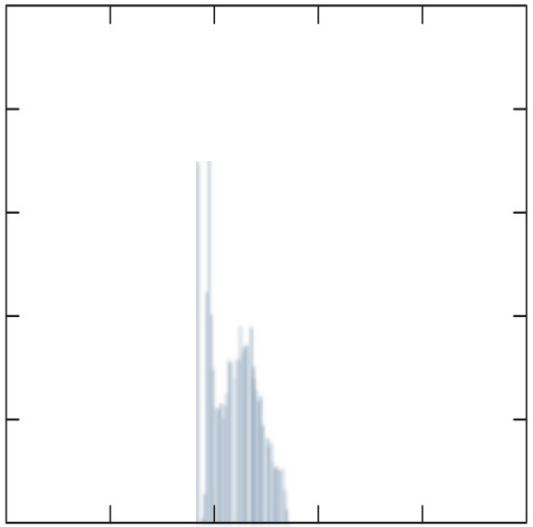


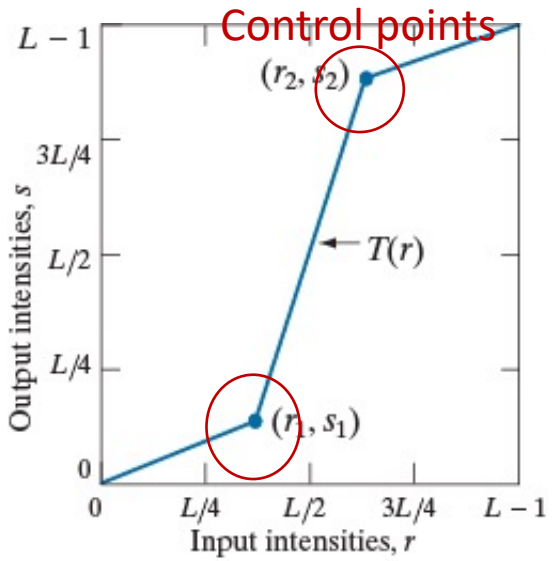

$s = \mathrm{T}(r) = ?$

- $r \le r_1$
  - $s =$
- $r_1 \le r < r_2$
  - $s =$
- $r_2 \le r$
  - $s =$

# Contrast Stretching





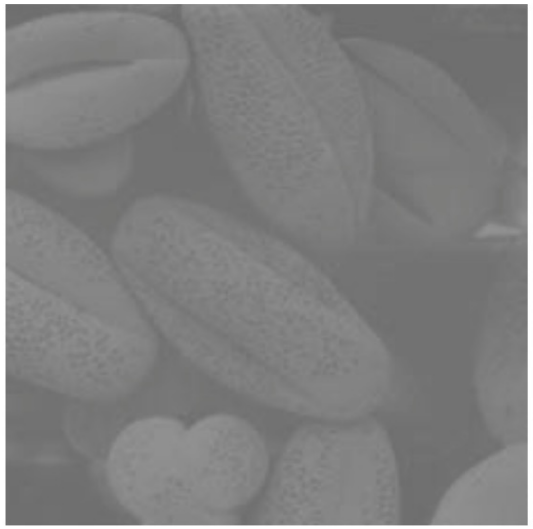


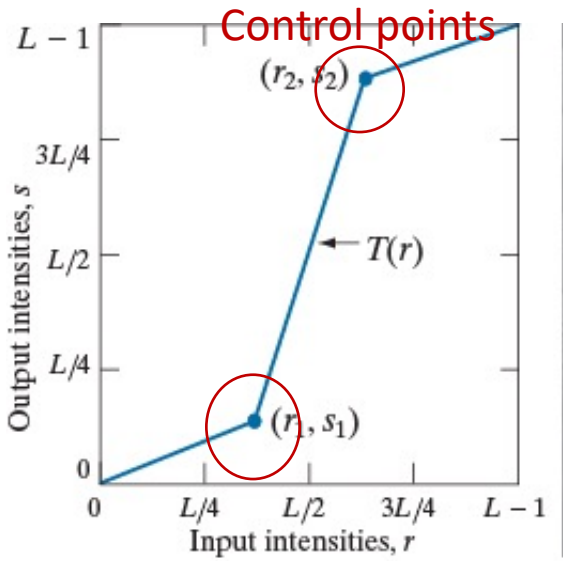

$s = \mathrm{T}(r) = ?$

- $r \leq r_1$
  - $s = s_1 . \frac{r}{r_1}$
- $r_1 \leq r < r_2$
  - $s = (s_2 - s_1) . \frac{r - r_1}{r_2 - r_1} + s_1$
- $r_2 \leq r$
  - $s = (L - 1 - s_2) . \frac{r - r_2}{L - 1 - r_2} + s_2$

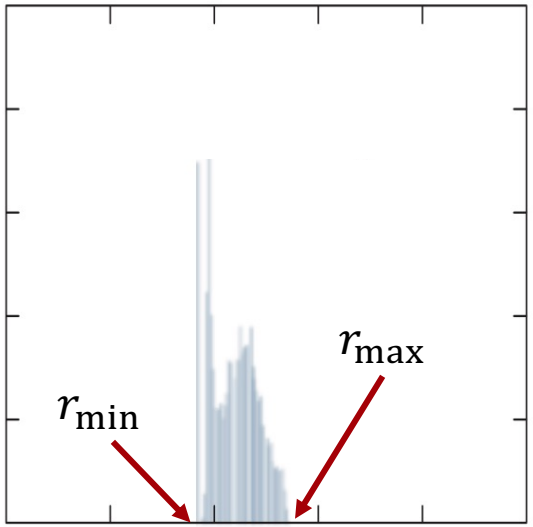
# Contrast Stretching





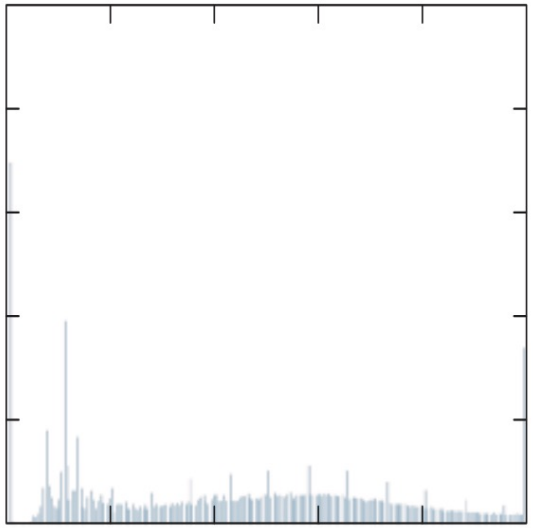





Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times.

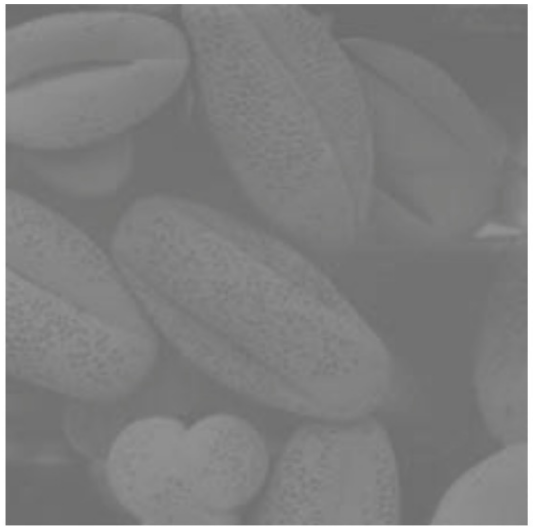
(Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

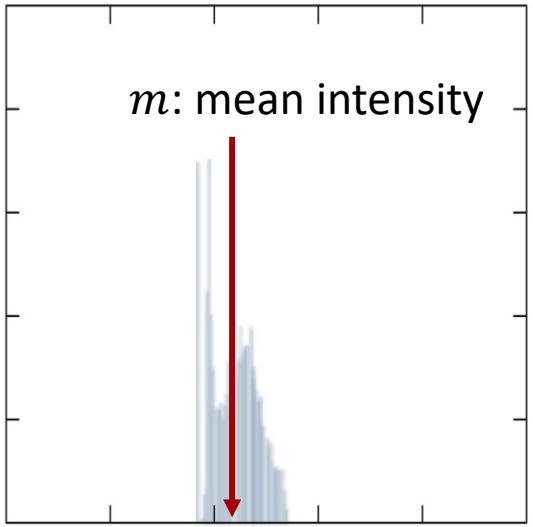


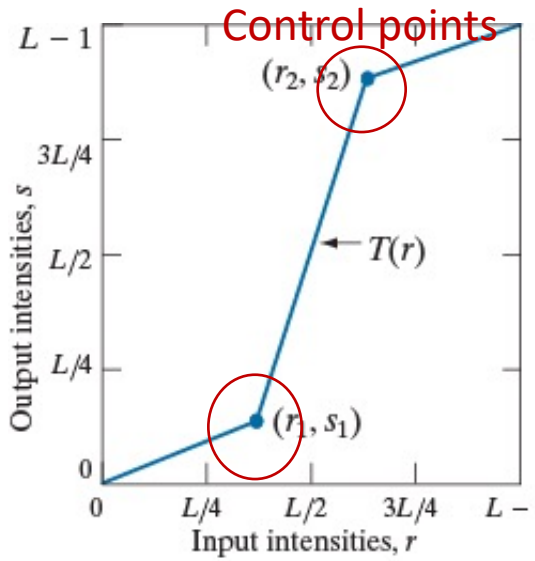
- Parameters:
  - $(r_1, s_1) = (r_{\min}, 0)$
  - $(r_2, s_2) = (r_{\max}, L-1)$

# Contrast Stretching: Extreme case - Thresholding








Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times.

(Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

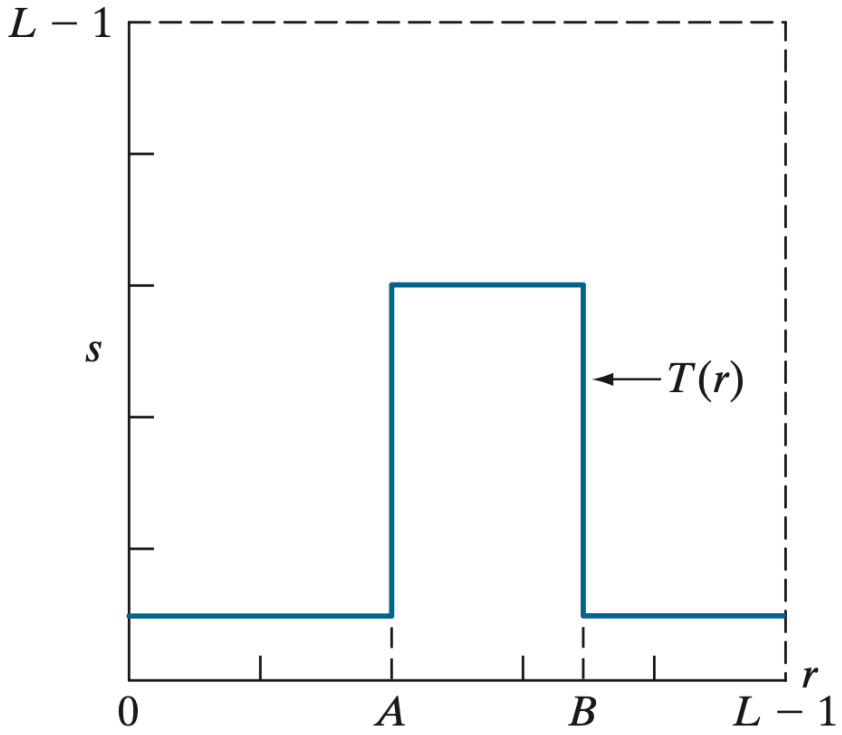


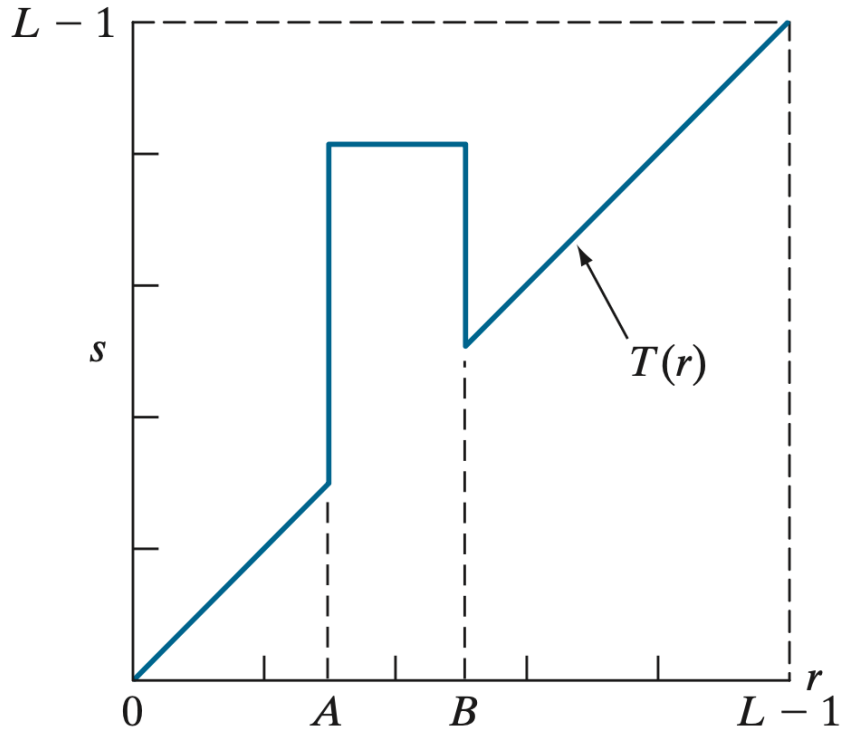
- Parameters:
  - $(r_1, s_1) = (m, 0)$
  - $(r_2, s_2) = (m, L-1)$

# Intensity-level slicing

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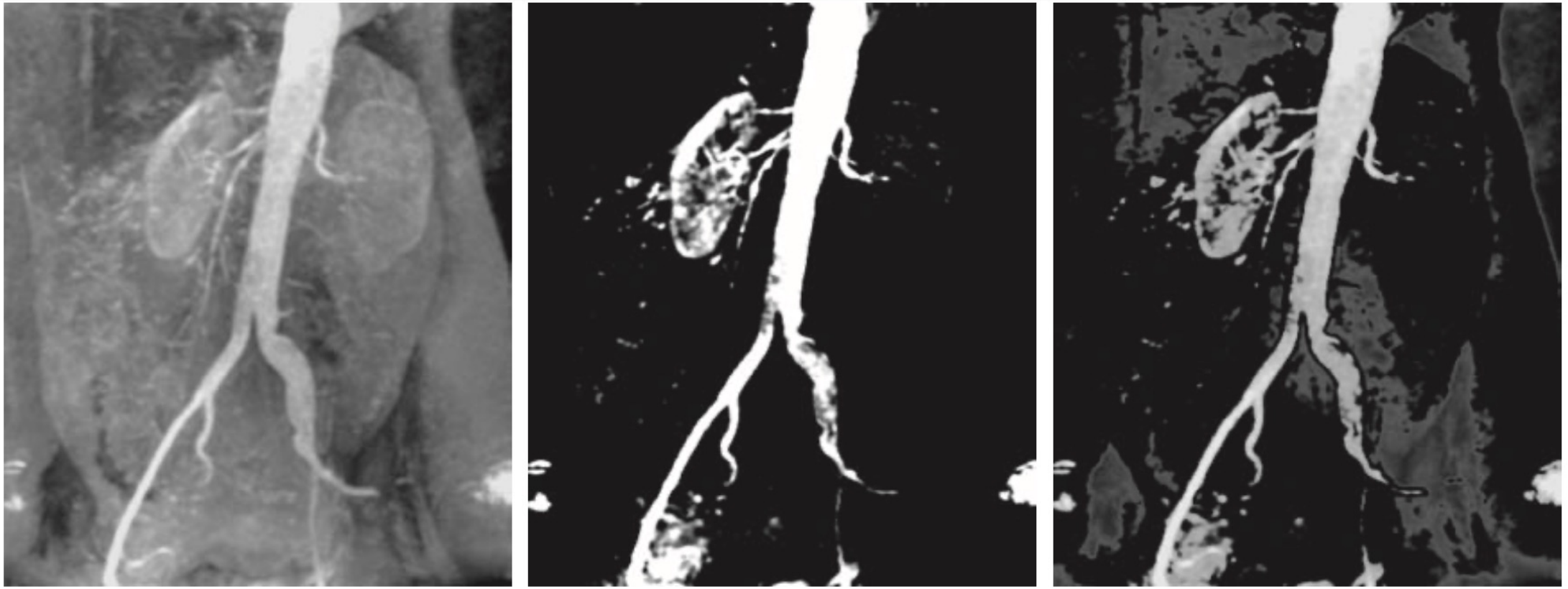
- What's the function of the transform $s = T(r)$ in the figures below?



# Intensity-level slicing





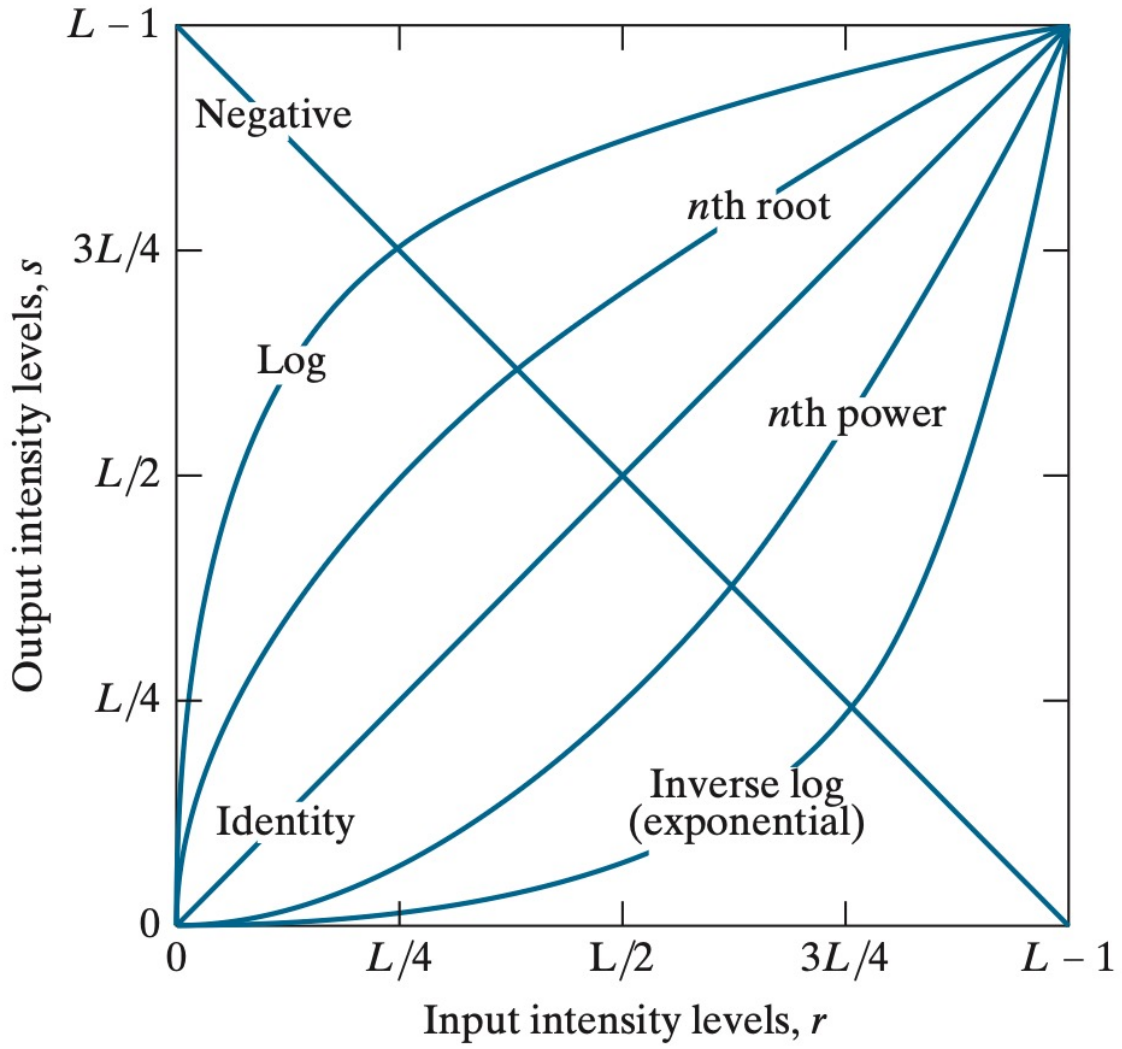
a b c

**FIGURE 3.12** (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected range set near black, so that the grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

# Log Transformation

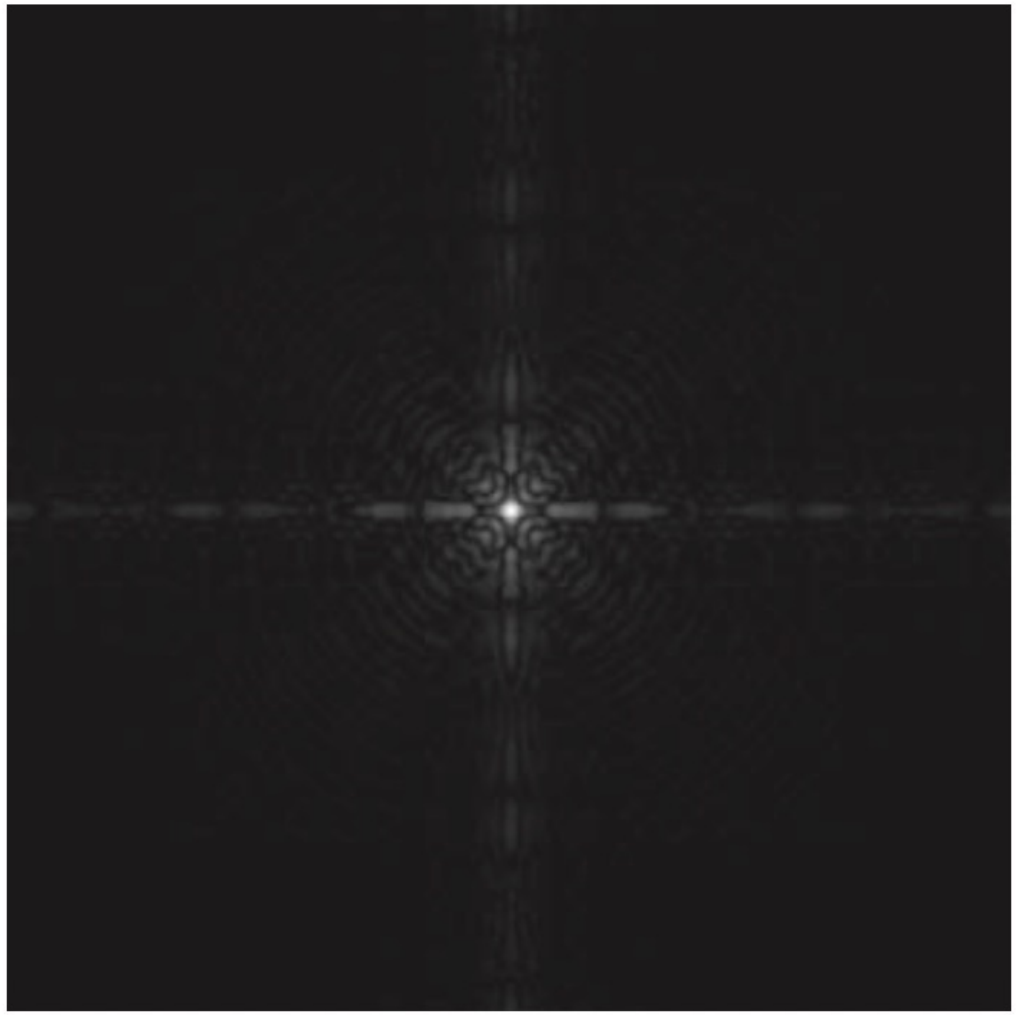


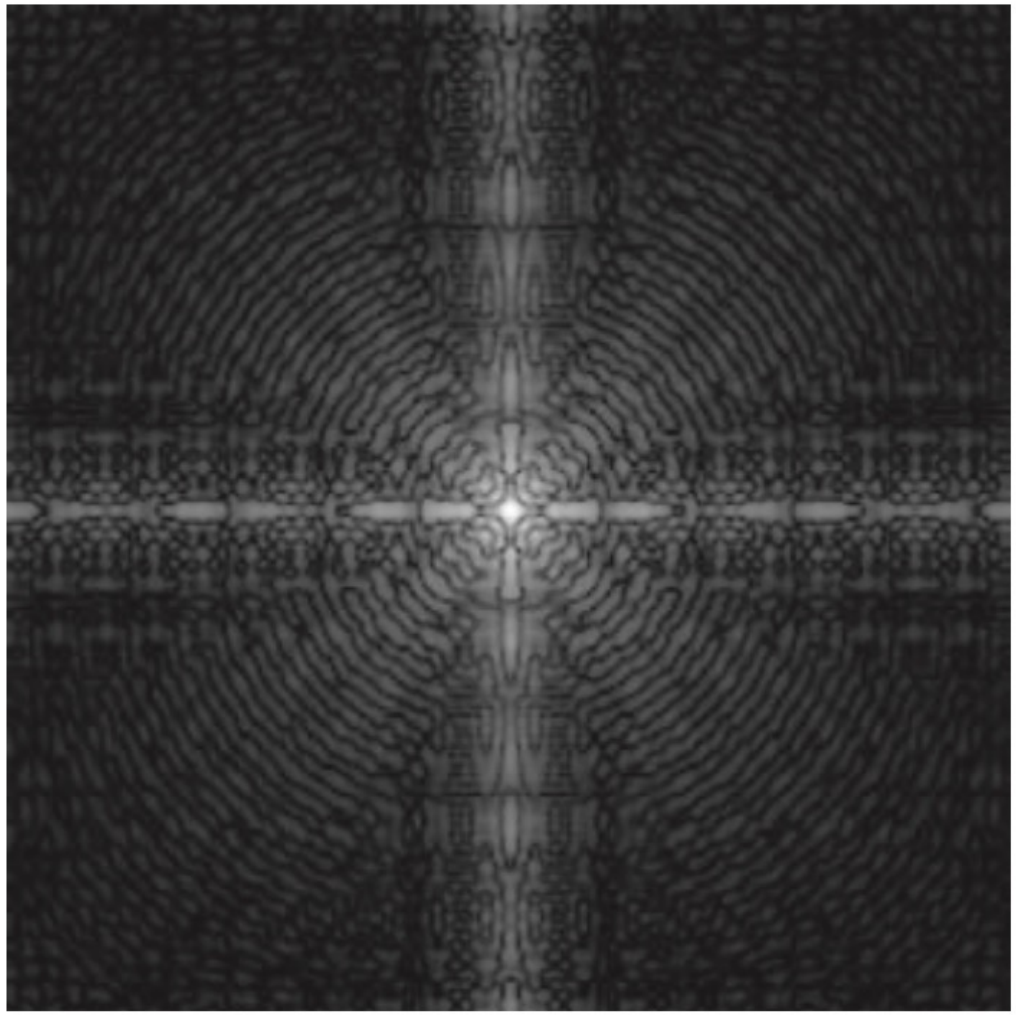
- **Log Transformation （对数变换）**
  - $s = c\ln(1 + r)$
- **Inverse Log Transformation（反对数变换）**
  - $s = c' \cdot e^r - 1$



# Fourier Spectrum(log trans)







# Gamma Transformation (power law)

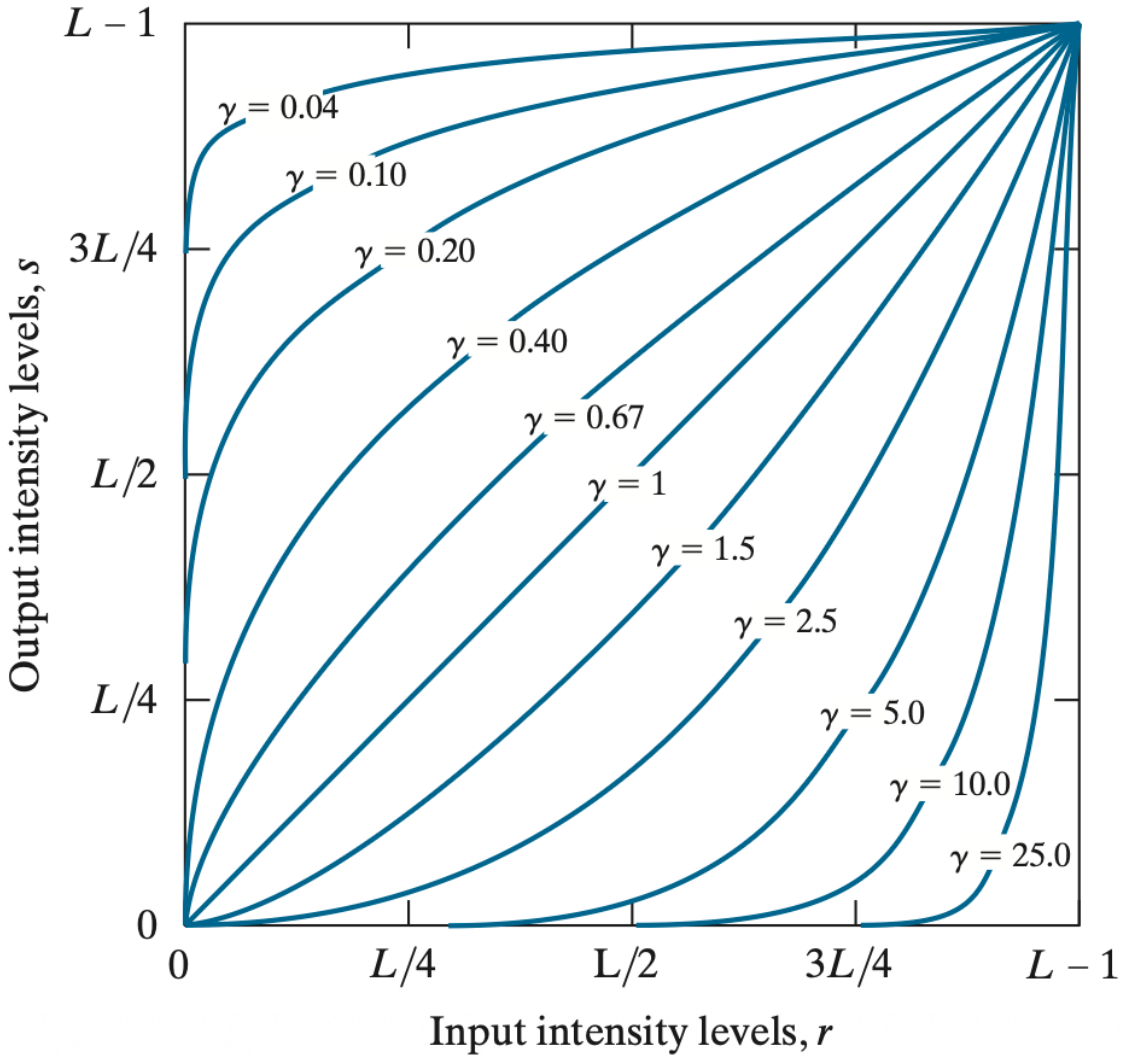
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- **Gamma Transformation**
  （伽马变换）

$$s = c \cdot r^{\gamma}$$

**or**

$$s = c \cdot (r + \varepsilon)^{\gamma}$$



# Gamma Transformation ($\gamma$<1): expansion of intensity levels







Fractured spine



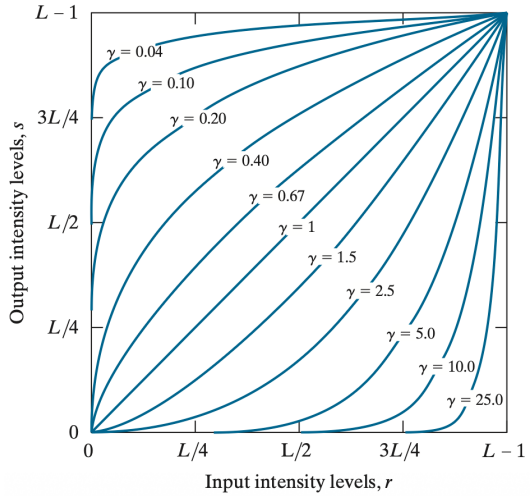
$\gamma = 0.6$



$\gamma = 0.4$



$\gamma = 0.3$



# Gamma Transformation ($\gamma$ >1): compression of intensity levels

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Aerial image



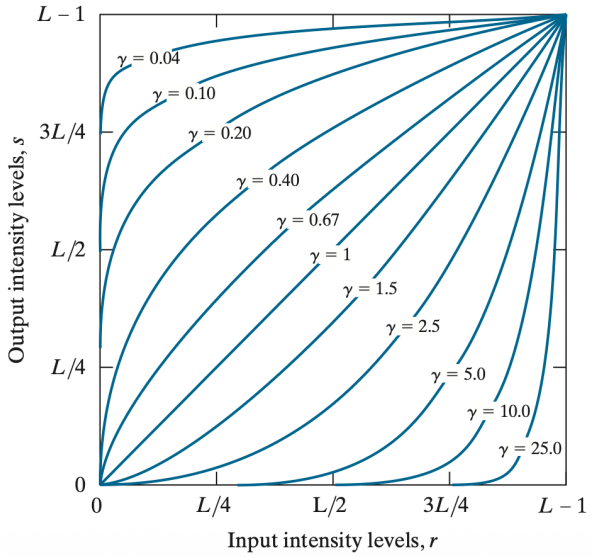
$\gamma = 3.0$



$\gamma = 4.0$



$\gamma = 5.0$




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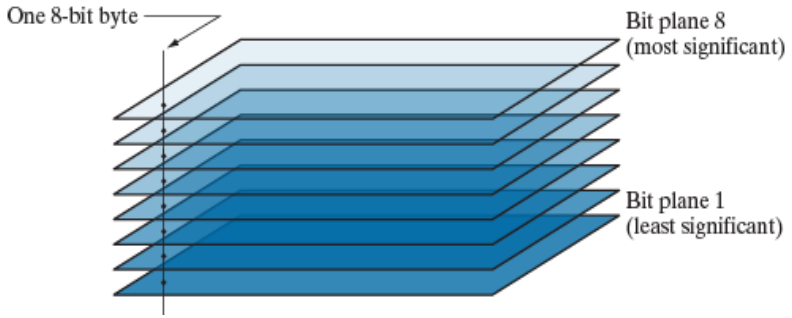
# Bit-Plane Slicing

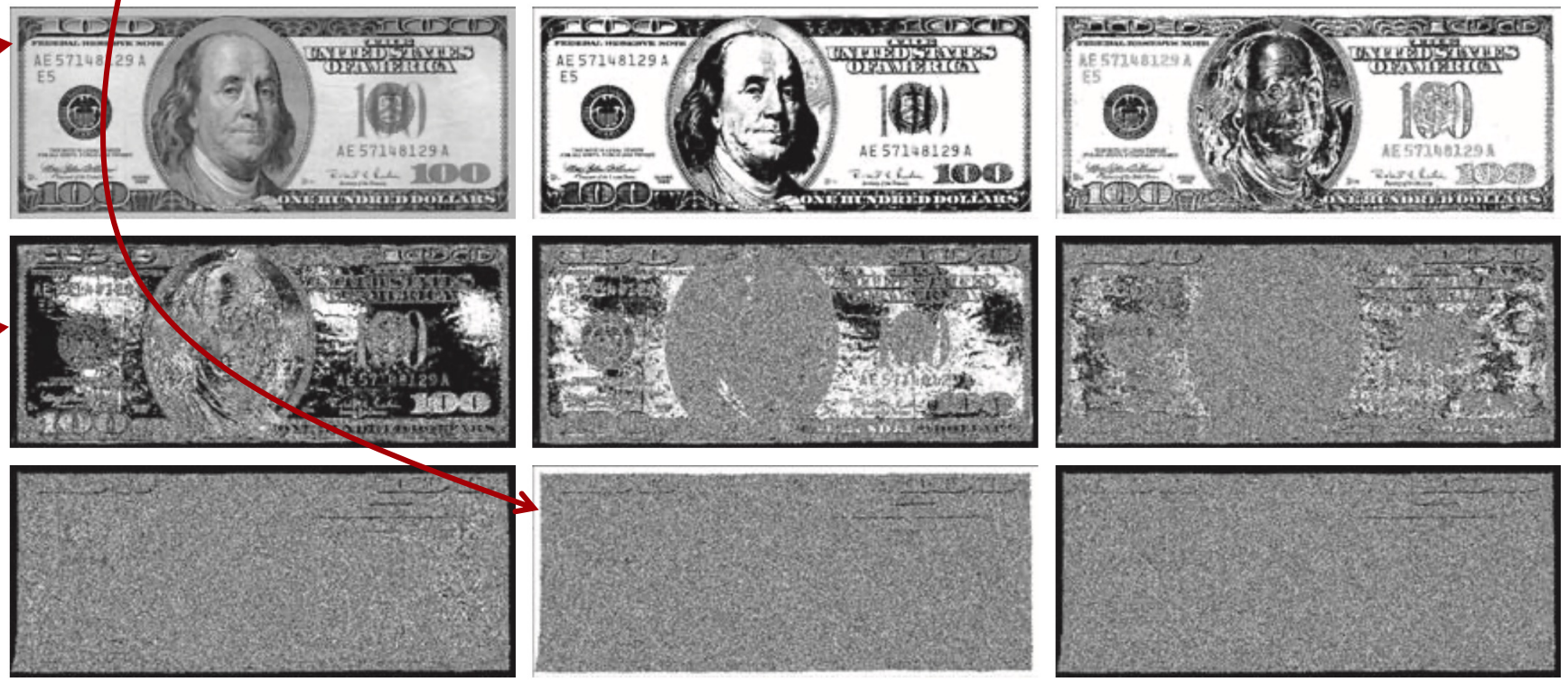


194d = 0b 1 1 0 0 0 0 1 0

bit 8

bit 1



a b c
d e f
g h i

**FIGURE 3.14** (a) An 8-bit gray-scale image of size $550 \times 1192$ pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image..

# Bit-Plane Slicing

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- Useful in image compression



Original image



bit planes 8 and 7



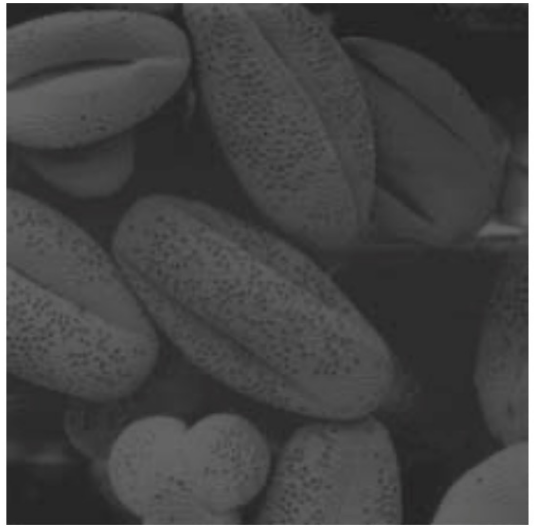
bit planes 8, 7 and 6

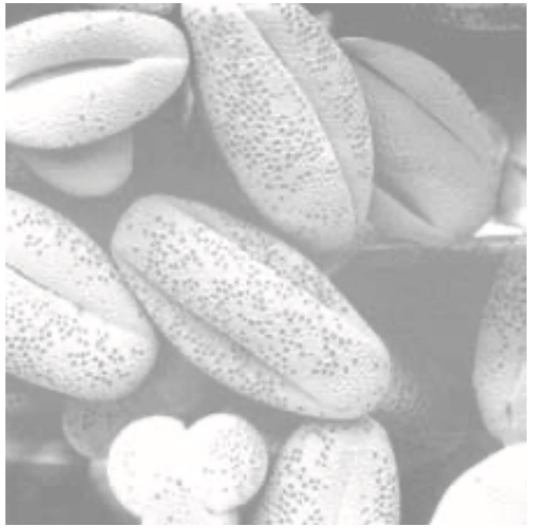


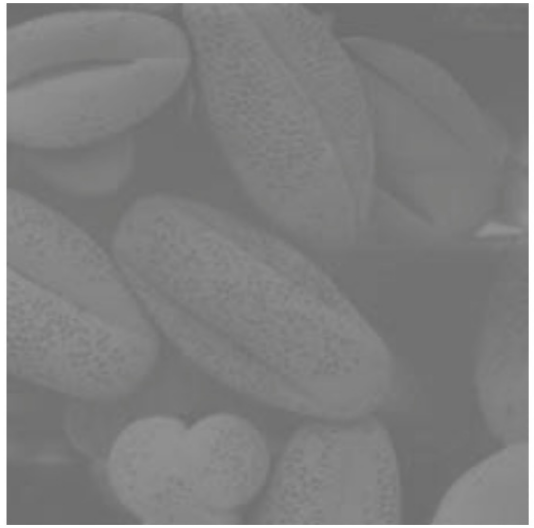
bit planes 8, 7, 6 and 5

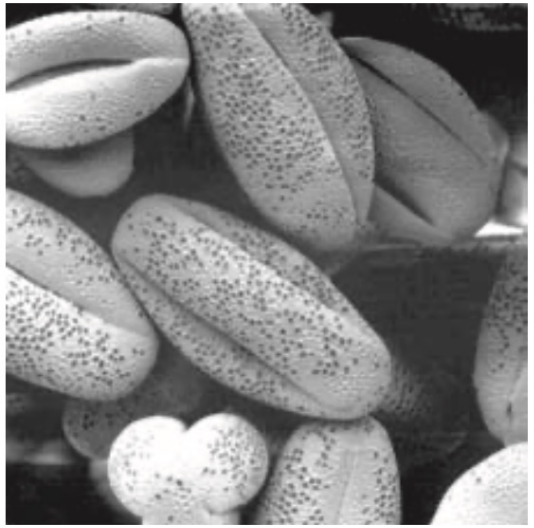
# Manipulating histograms

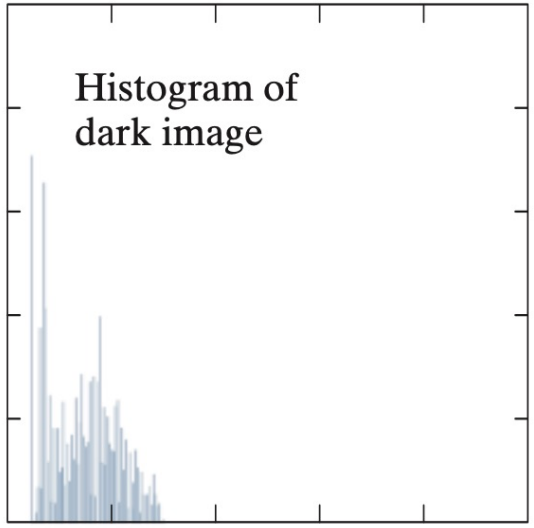


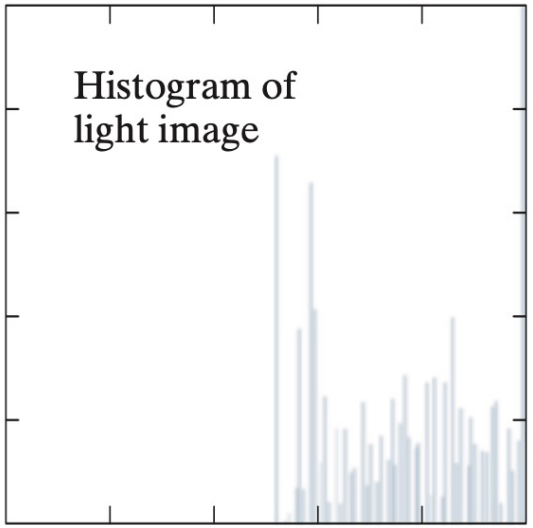


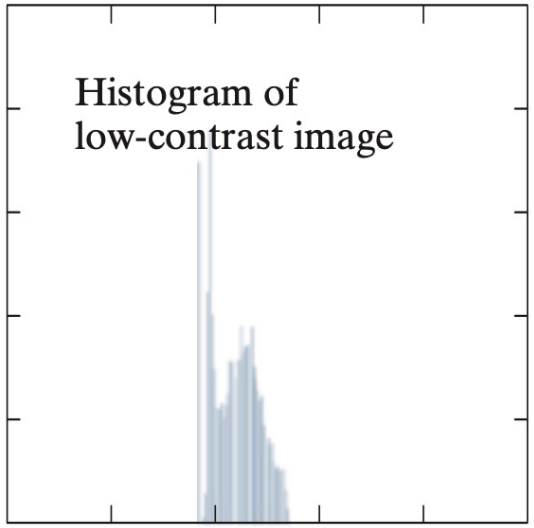


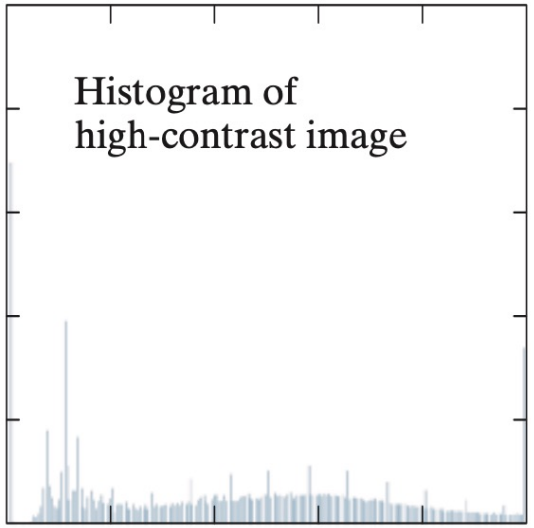






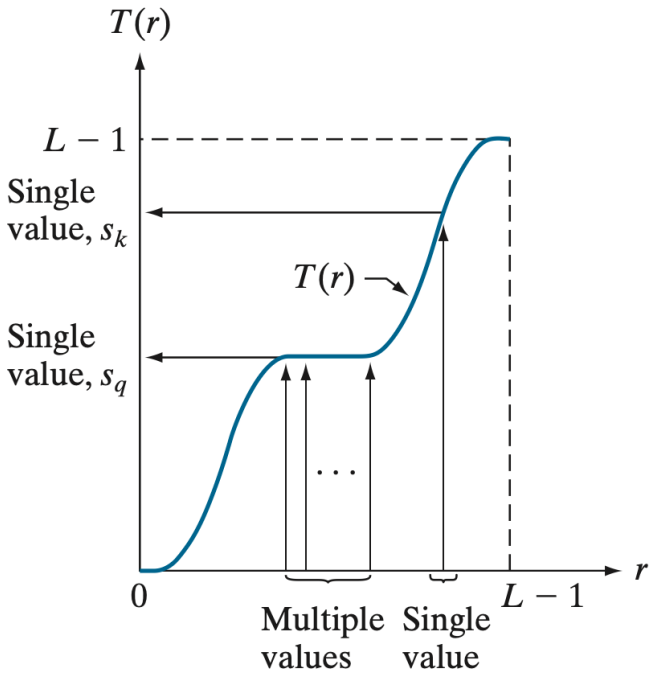



# Basis of Histogram Processing

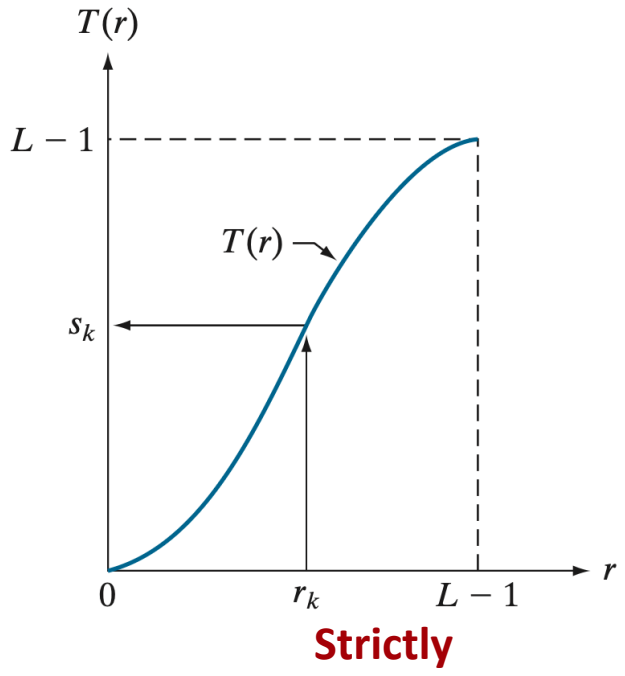


- Given intensity transformation $s = T(r)$, where:
  - $T(r)$ is strictly monotonically increasing function （严格单调递增函数)

    in the interval $0 \le r \le L-1 \rightarrow T(r_2) > T(r_1)$ if $r_2 > r_1$
  - $0 \le T(r) \le L-1$ for $0 \le r \le L-1$
- T is then a “one-to-one” map
- The inverse transform exists
  - $r = T^{-1}(s)$




Monotonically increasing

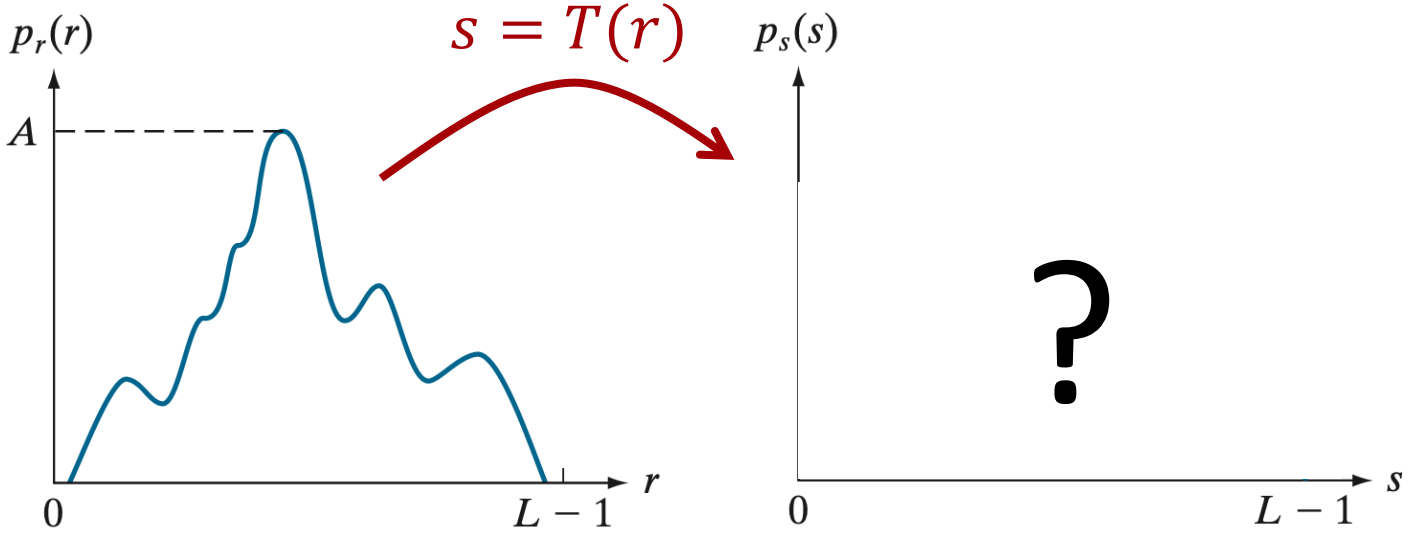



**Strictly** monotonically increasing

# Histogram Equalization

上海科技大学
ShanghaiTech University

- We see intensity of an image as a density probability functions $p_r(r)$ and $p_s(s)$ in $[0, L-1]$, with transform $s = T(r)$




- Probability theory gives us $\boldsymbol{p_s(s) = p_r(r) \cdot \left|\frac{dr}{ds}\right|}$, if $T(r)$ continuous and differentiable
- Let's look at the transform

$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$

# Histogram Equalization



- Let's look at the transform

$$s = T(r) = (L-1)\int_0^r p_r(w)dw$$

- Integral part = *cumulative distribution function* (CDF) of random variable r.
  - This is the area under a curve
  - This is always increasing when r increases
  - T is strictly monotonically increasing (if all intensity values r are in the image)

- When $r = L-1$, the integral equals to 1 because it is a *PDF*
  - So, $0 \leq T(r) \leq L-1$ for $0 \leq r \leq L-1$

- Our conditions are verified !

# Histogram Equalization



- $\frac{ds}{dr} = \frac{dT(r)}{dr}$

$$= (L-1)\frac{d}{dr}\left[\int_0^r p_r(w)dw\right]$$

$s = T(r) = (L-1)\int_0^r p_r(w)dw$

Leibniz's rule

$$= (L-1)p_r(r)$$

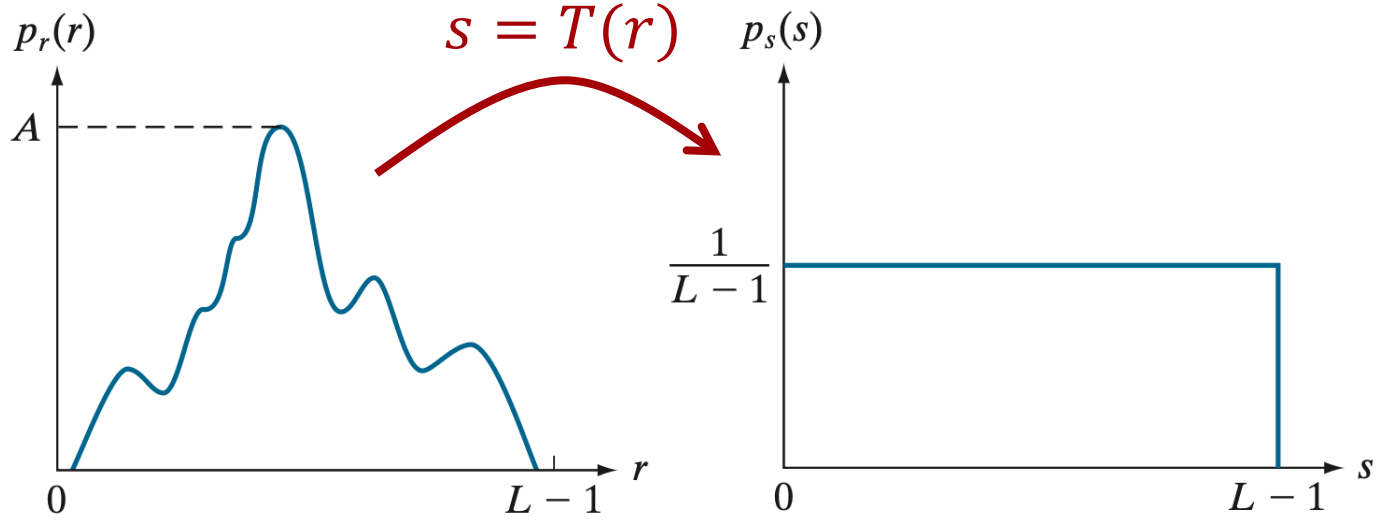
From probabilities

- $p_s(s) = p_r(r) \cdot \left|\frac{dr}{ds}\right|$

$$= p_r(r) \cdot \left|\frac{1}{(L-1)p_r(r)}\right|$$

Always >= 0

- $p_s(s) = \frac{1}{L-1}, 0 \le s \le L-1$



# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ |
|---|---|---|---|
| 0 | 790 | | |
| 1 | 1023 | | |
| 2 | 850 | | |
| 3 | 656 | | |
| 4 | 329 | | |
| 5 | 245 | | |
| 6 | 122 | | |
| 7 | 81 | | |

# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ |
|---|---|---|---|
| 0 | 790 | 0.19 | |
| 1 | 1023 | 0.25 | |
| 2 | 850 | 0.21 | |
| 3 | 656 | 0.16 | |
| 4 | 329 | 0.08 | |
| 5 | 245 | 0.06 | |
| 6 | 122 | 0.03 | |
| 7 | 81 | 0.02 | |

# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ |
|---|---|---|---|
| 0 | 790 | 0.19 | 1.33 |
| 1 | 1023 | 0.25 | 3.08 |
| 2 | 850 | 0.21 | 4.55 |
| 3 | 656 | 0.16 | 5.67 |
| 4 | 329 | 0.08 | 6.23 |
| 5 | 245 | 0.06 | 6.65 |
| 6 | 122 | 0.03 | 6.86 |
| 7 | 81 | 0.02 | 7.00 |

must be an integer !

$$s_0 = T(r_0) = 7.\sum_{j=0}^{0} p_r(r_j) = 7.p_r(r_0) = 1.33$$

$$s_1 = T(r_1) = 7.\sum_{j=0}^{1} p_r(r_j) = 7.\left(p_r(r_0) + p_r(r_1)\right) = 3.08$$

# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ | |
|---|---|---|---|---|
| 0 | 790 | 0.19 | 1.33 | 1 |
| 1 | 1023 | 0.25 | 3.08 | 3 |
| 2 | 850 | 0.21 | 4.55 | 5 |
| 3 | 656 | 0.16 | 5.67 | 6 |
| 4 | 329 | 0.08 | 6.23 | 6 |
| 5 | 245 | 0.06 | 6.65 | 7 |
| 6 | 122 | 0.03 | 6.86 | 7 |
| 7 | 81 | 0.02 | 7.00 | 7 |

# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ | | $s'_k$ | $n'_k$ | $p_s(s_k)$ |
|---|---|---|---|---|---|---|---|
| 0 | 790 | 0.19 | 1.33 | 1 | 0 | | |
| 1 | 1023 | 0.25 | 3.08 | 3 | 1 | | |
| 2 | 850 | 0.21 | 4.55 | 5 | 2 | | |
| 3 | 656 | 0.16 | 5.67 | 6 | 3 | | |
| 4 | 329 | 0.08 | 6.23 | 6 | 4 | | |
| 5 | 245 | 0.06 | 6.65 | 7 | 5 | | |
| 6 | 122 | 0.03 | 6.86 | 7 | 6 | | |
| 7 | 81 | 0.02 | 7.00 | 7 | 7 | | |

# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ | | $s'_k$ | $n'_k$ | $p_s(s_k)$ |
|---|---|---|---|---|---|---|---|
| 0 | 790 | 0.19 | 1.33 | 1 | 0 | 0 | |
| 1 | 1023 | 0.25 | 3.08 | 3 | 1 | 790 | |
| 2 | 850 | 0.21 | 4.55 | 5 | 2 | 0 | |
| 3 | 656 | 0.16 | 5.67 | 6 | 3 | 1023 | |
| 4 | 329 | 0.08 | 6.23 | 6 | 4 | 0 | |
| 5 | 245 | 0.06 | 6.65 | 7 | 5 | 850 | |
| 6 | 122 | 0.03 | 6.86 | 7 | 6 | 985 | |
| 7 | 81 | 0.02 | 7.00 | 7 | 7 | 448 | |

# Histogram Equalization – Numerical example



- 3-bit image (L=8)
- Size 64x64 pixels (M.N=4096)
- Intensity levels are integers in the range: $[0, L-1] = [0, 7]$

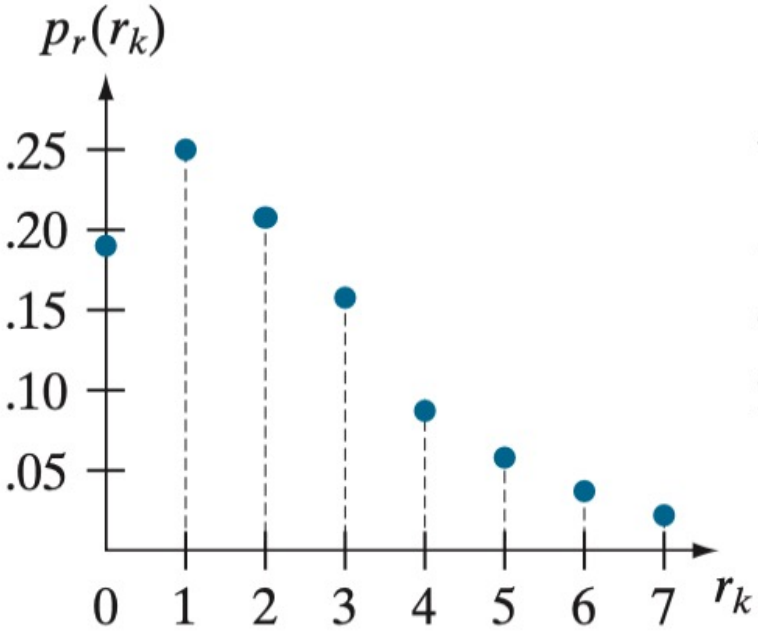
$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

$$k = 0,1,2,\ldots,L-1, k \le r$$

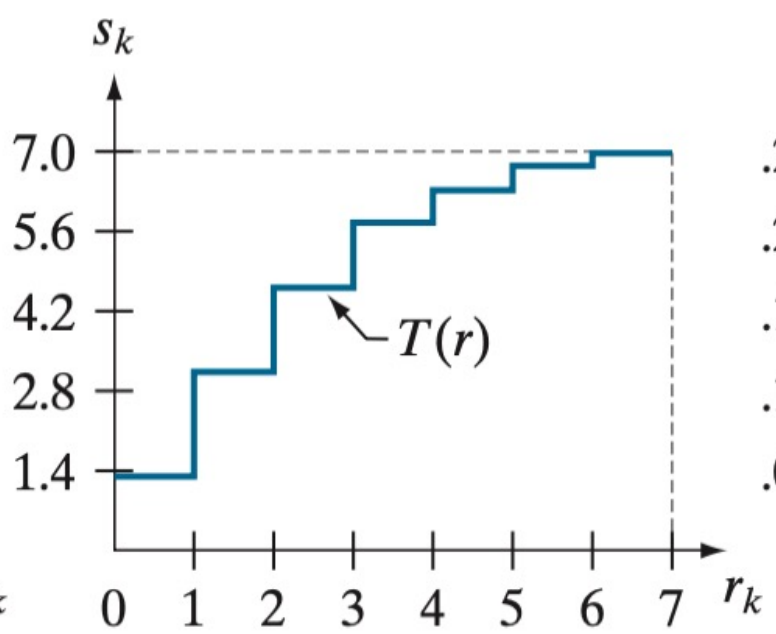
| $r_k$ | $n_k$ | $p_r(r_k)=n_k/MN$ | $s_k$ | | $s'_k$ | $n'_k$ | $p_s(s_k)$ |
|---|---|---|---|---|---|---|---|
| 0 | 790 | 0.19 | 1.33 | 1 | 0 | 0 | 0 |
| 1 | 1023 | 0.25 | 3.08 | 3 | 1 | 790 | 0.19 |
| 2 | 850 | 0.21 | 4.55 | 5 | 2 | 0 | 0 |
| 3 | 656 | 0.16 | 5.67 | 6 | 3 | 1023 | 0.25 |
| 4 | 329 | 0.08 | 6.23 | 6 | 4 | 0 | 0 |
| 5 | 245 | 0.06 | 6.65 | 7 | 5 | 850 | 0.21 |
| 6 | 122 | 0.03 | 6.86 | 7 | 6 | 985 | 0.24 |
| 7 | 81 | 0.02 | 7.00 | 7 | 7 | 448 | 0.11 |

# Histogram Equalization – Numerical example

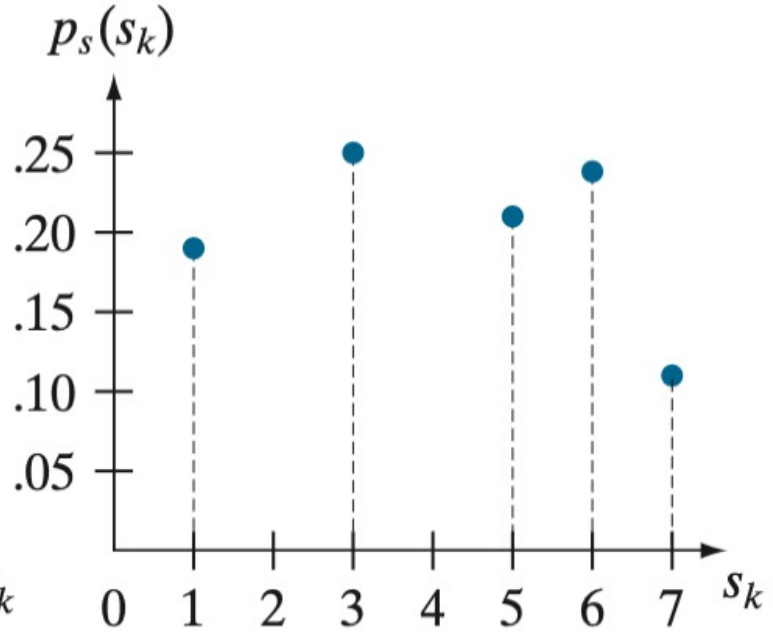





Original histogram




Transformation function

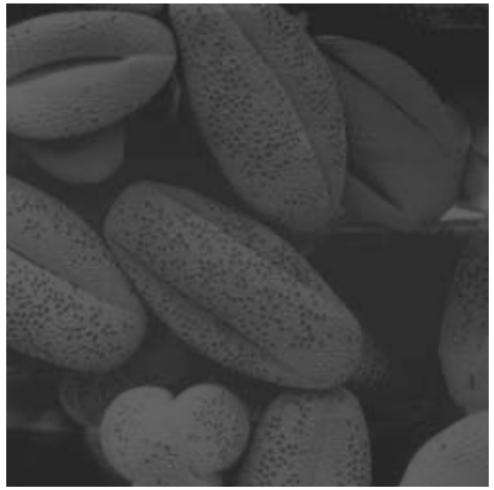


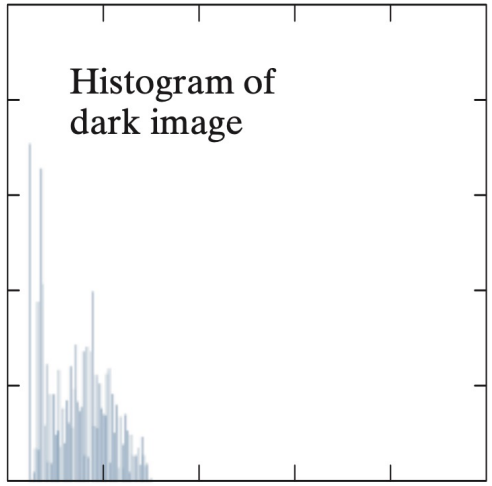

Equalized histogram

# Histogram Equalization – Example images

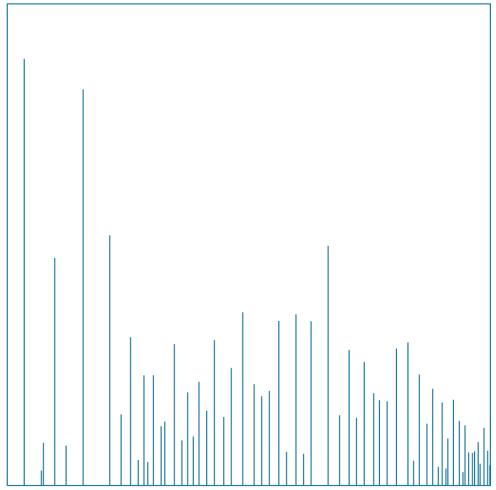


1

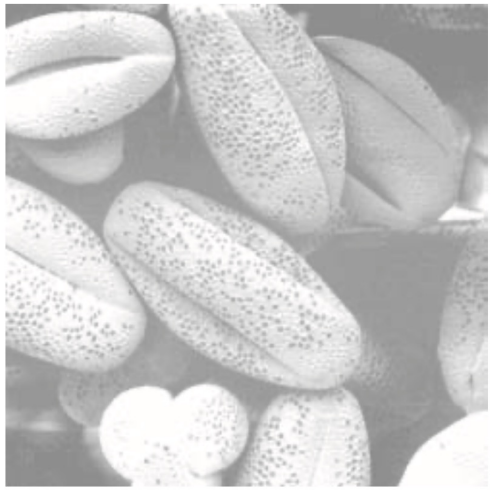


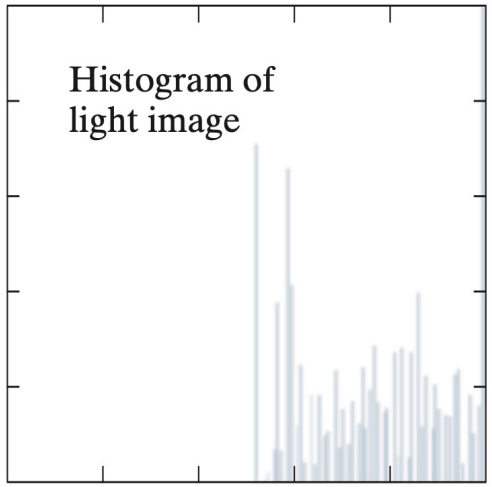


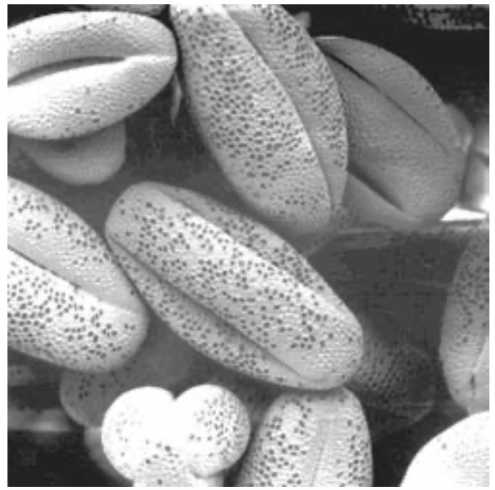


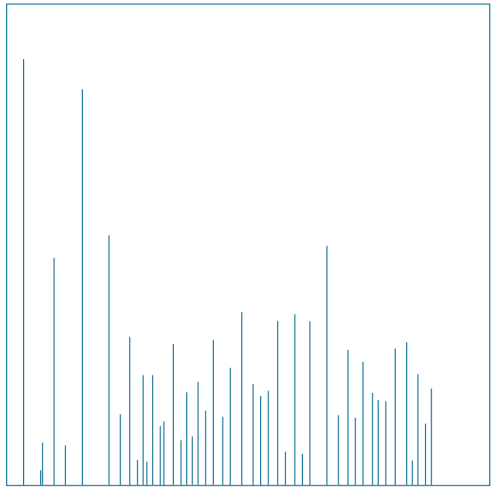



2





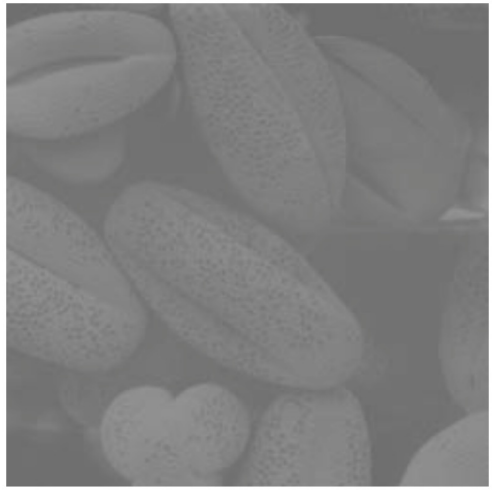



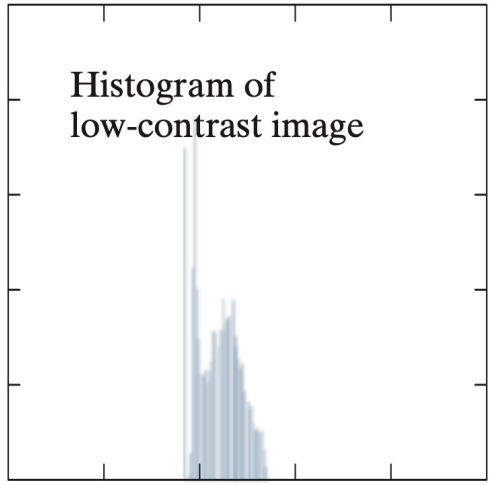


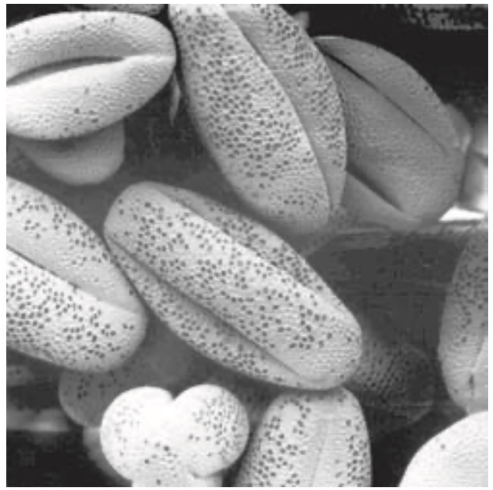
# Histogram Equalization – Example images

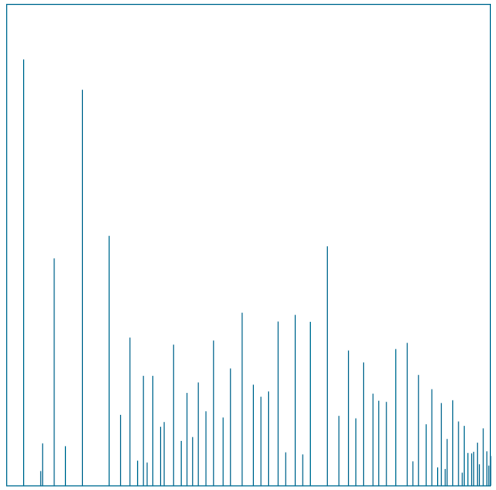


3

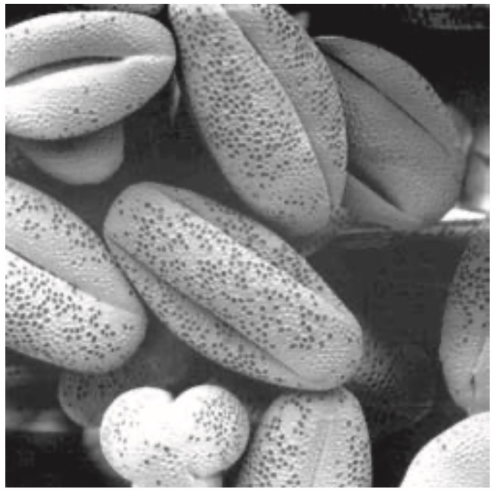


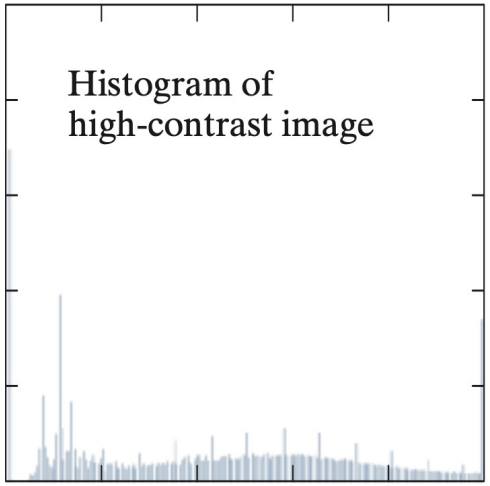


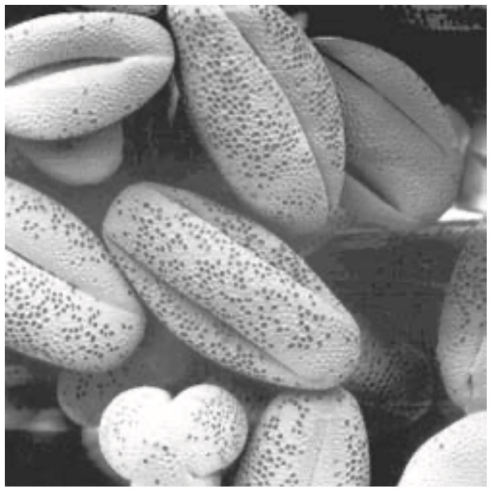



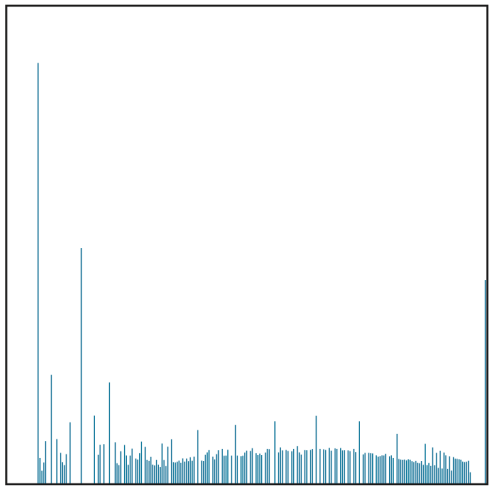


4



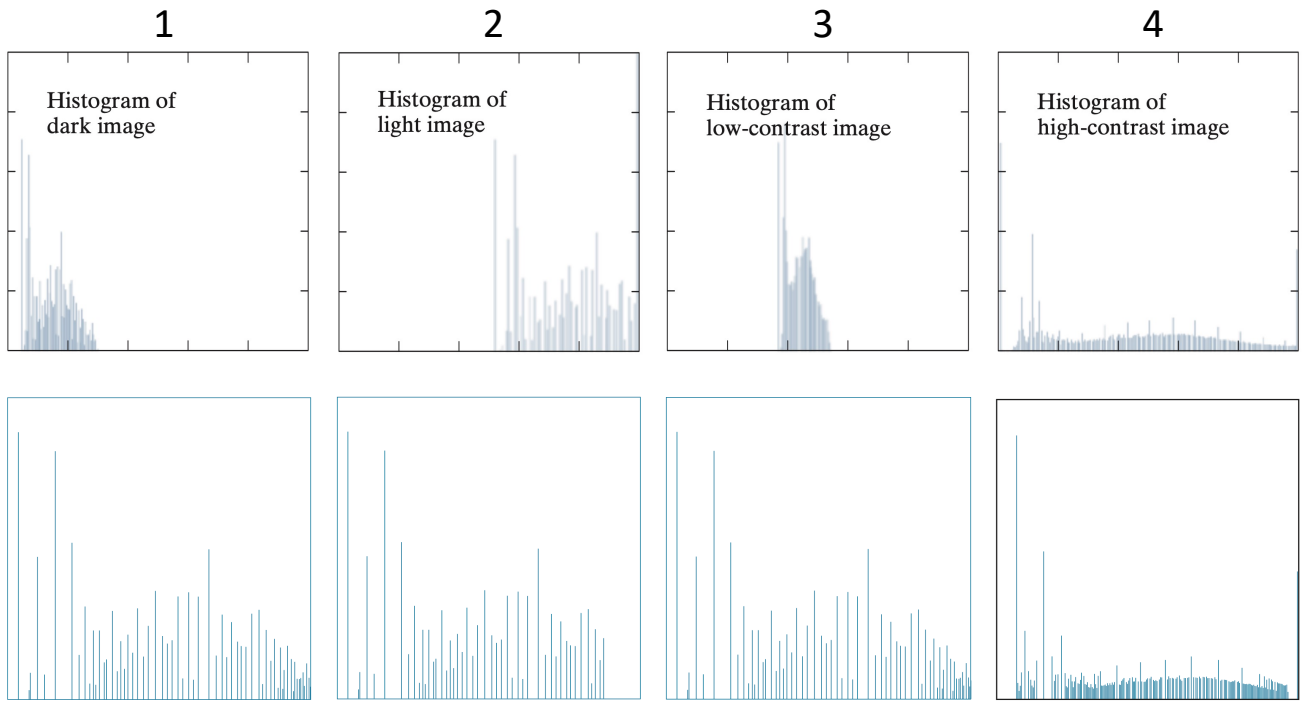


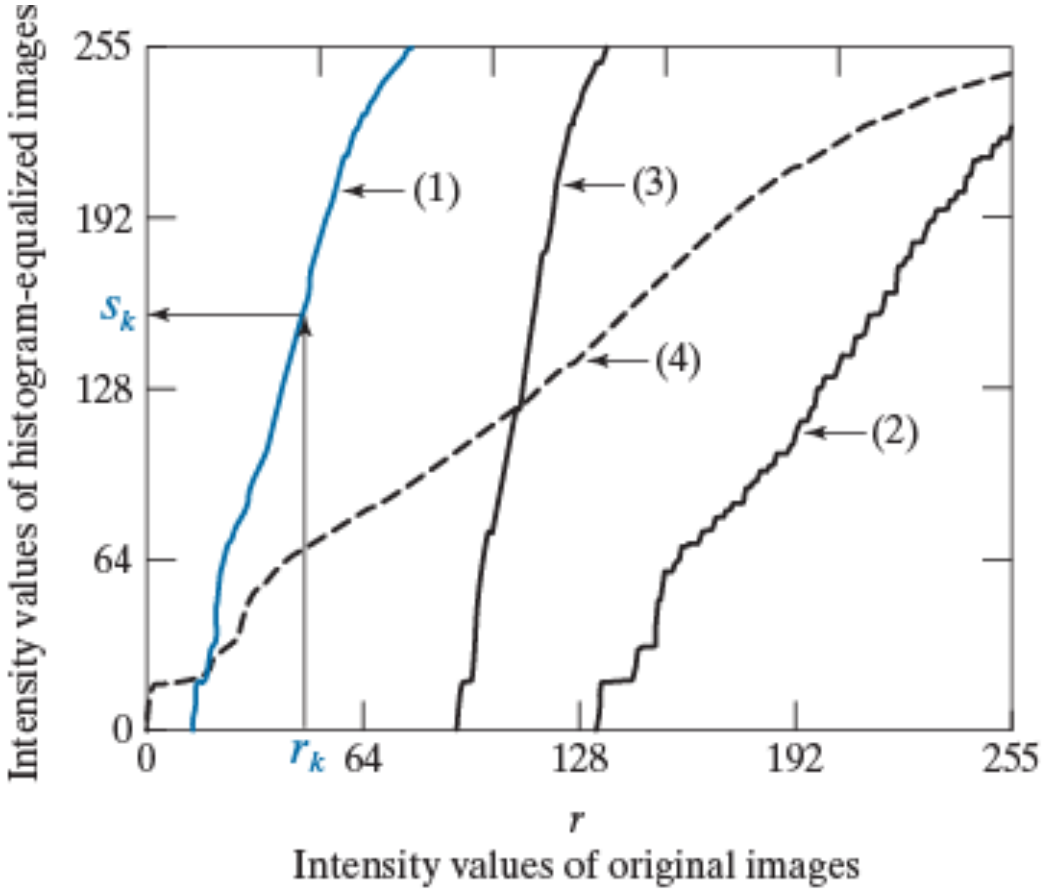





# Histogram Equalization – Example images: mapping function



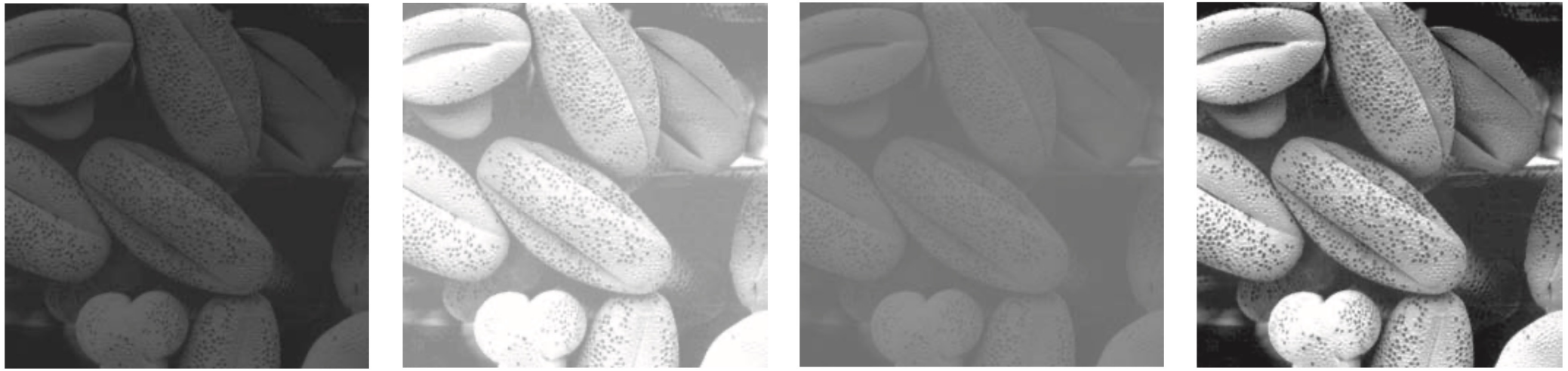


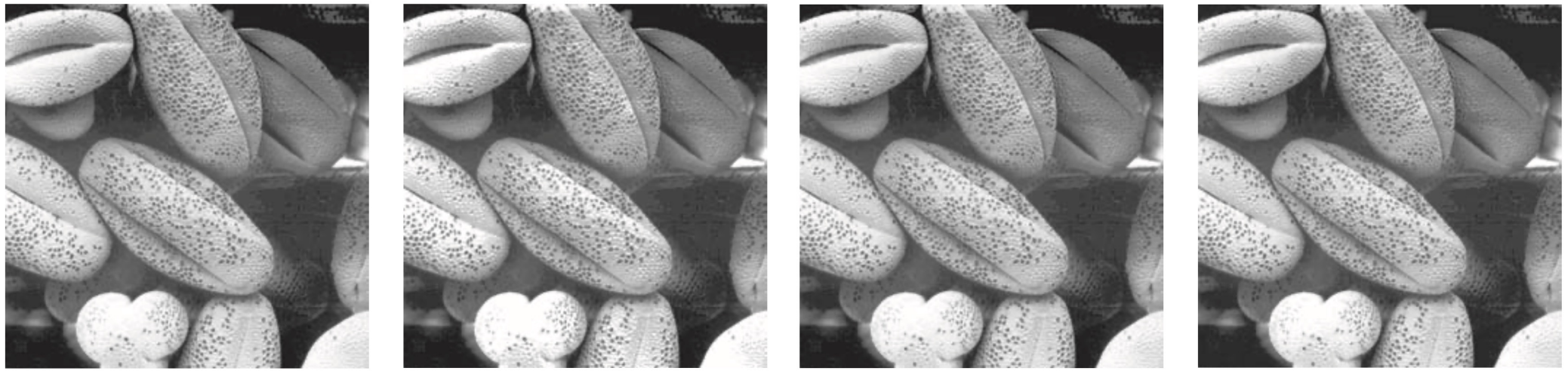
# Histogram Equalization – A good automatic method



input



output



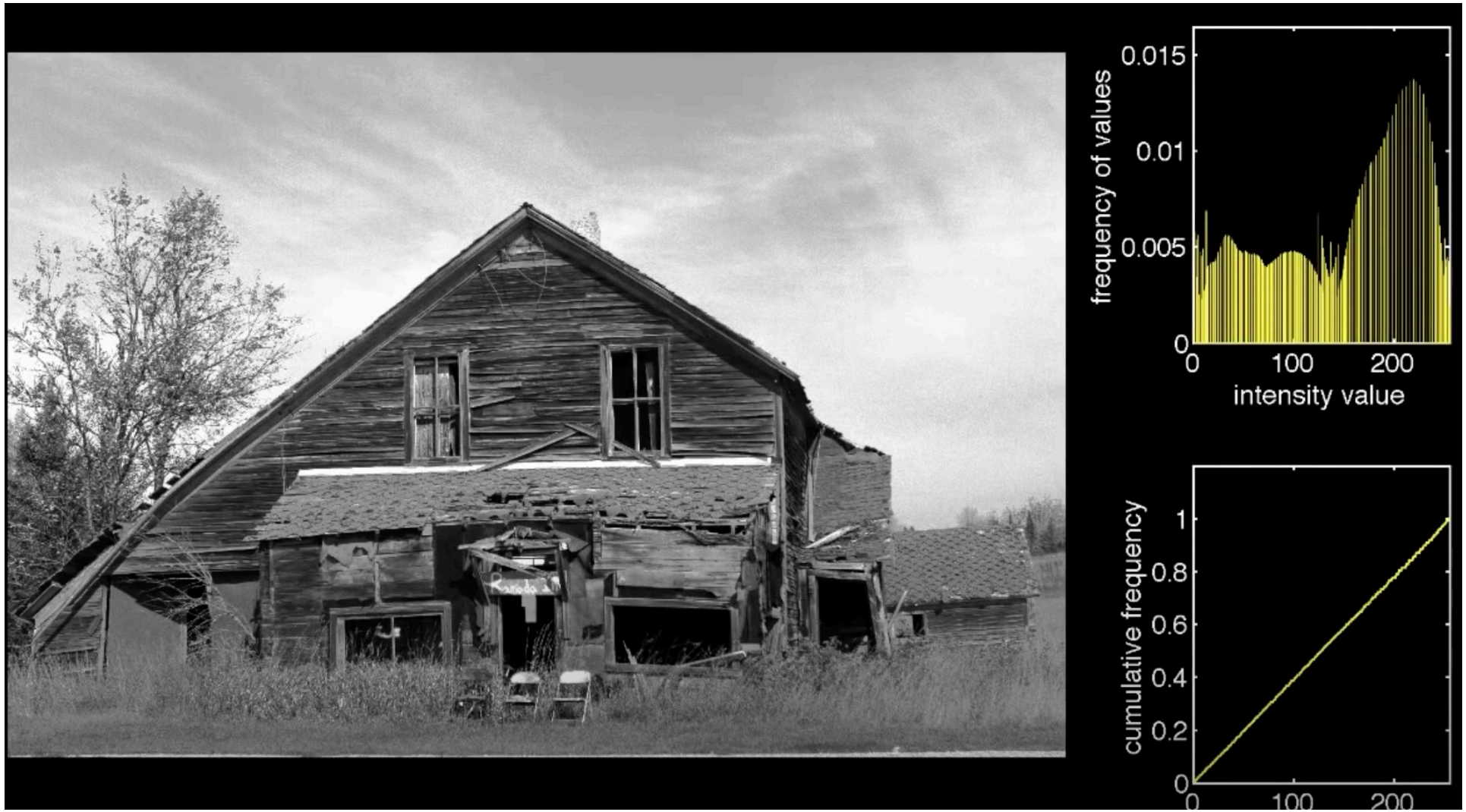
# Histogram Equalization – More examples





# Histogram Equalization – More examples





# Histogram Equalization – More examples





# Histogram Equalization – More examples





# Histogram Matching or Histogram Specification



- **Generate a processed image with a specified histogram:**

Estimated from input image

For input : $s = T(r) = (L-1)\int_0^r p_r(w)dw$

given　　Input and Output are equalized !

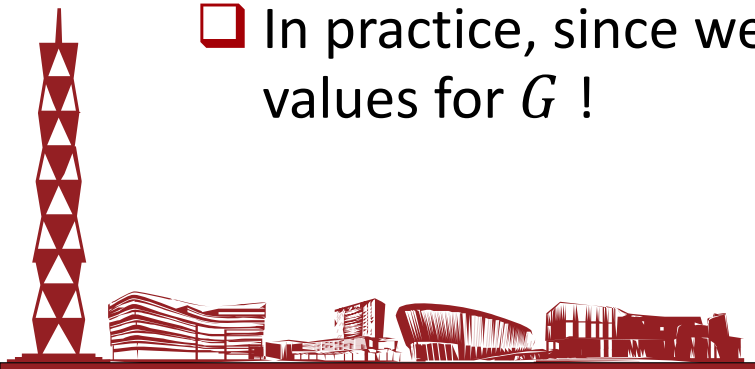
For output : $G(z) = (L-1)\int_0^z p_z(t)dt = s$

Therefore　　$z = G^{-1}(s) = G^{-1}[T(r)]$

# Histogram Matching or Histogram Specification



- Steps
  - Compute histogram $p_r(r)$ from the input image
  - Use the specified PDF, $p_z(z)$ to obtain $G(z)$
  - Compute the inverse transformation $z = G^{-1}(s)$, which is a mapping from $s$ to $z$
  - Equalize the input image to obtain the $s$ values, apply mapping $z = G^{-1}(s)$
- In practice, since we are using integer values, we only need to compute all possible values for $G$ !



# Histogram Matching or Histogram Specification



- Actual steps when working with histograms
  - Obtain $p_r(r)$ from the input image, equalize and round the $s_k$ values.
  - Compute all values of $G(z_q), q = 0, 1, \ldots, L-1$ where $p_z(z_i)$ are the values of the specified histogram. Round the values of $G$ to integers in the range $[0, L-1]$ and store in a lookup table.
  - Use this table to find the corresponding value of $z_q$ so that $G(z_q)$ is closest to $s_k$. Store the mapping from $s$ to $z$. When more than one value of $z_q$ gives the same match, choose the smallest by convention.
  - Use this mapping to transform the equalized pixels $s_k$ into the corresponding values $z_q$



# Histogram Matching – Numerical example

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- This is where we start from

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | | 0 | 0 | | | | | |
| 1 | 0.25 | | 1 | 0 | | | | | |
| 2 | 0.21 | | 2 | 0 | | | | | |
| 3 | 0.16 | | 3 | 0.15 | | | | | |
| 4 | 0.08 | | 4 | 0.20 | | | | | |
| 5 | 0.06 | | 5 | 0.30 | | | | | |
| 6 | 0.03 | | 6 | 0.20 | | | | | |
| 7 | 0.02 | | 7 | 0.15 | | | | | |

input　　specified　　actual (output)



# Histogram Matching – Numerical example



- Step 1: equalize and round the $s_k$ values from the input image

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | | 0 | 0 | | | | | |
| 1 | 0.25 | | 1 | 0 | | | | | |
| 2 | 0.21 | | 2 | 0 | | | | | |
| 3 | 0.16 | | 3 | 0.15 | | | | | |
| 4 | 0.08 | | 4 | 0.20 | | | | | |
| 5 | 0.06 | | 5 | 0.30 | | | | | |
| 6 | 0.03 | | 6 | 0.20 | | | | | |
| 7 | 0.02 | | 7 | 0.15 | | | | | |

input　　specified　　actual (output)

# Histogram Matching – Numerical example



| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | | | | | |
| 1 | 0.25 | 3 | 1 | 0 | | | | | |
| 2 | 0.21 | 5 | 2 | 0 | | | | | |
| 3 | 0.16 | 6 | 3 | 0.15 | | | | | |
| 4 | 0.08 | 6 | 4 | 0.20 | | | | | |
| 5 | 0.06 | 7 | 5 | 0.30 | | | | | |
| 6 | 0.03 | 7 | 6 | 0.20 | | | | | |
| 7 | 0.02 | 7 | 7 | 0.15 | | | | | |

input　　specified　　actual (output)

# Histogram Matching – Numerical example



- Step 2: Compute all values of $G(z_k), k = 0, 1, \ldots, L-1$ and round them

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | | | | | |
| 1 | 0.25 | 3 | 1 | 0 | | | | | |
| 2 | 0.21 | 5 | 2 | 0 | | | | | |
| 3 | 0.16 | 6 | 3 | 0.15 | | | | | |
| 4 | 0.08 | 6 | 4 | 0.20 | | | | | |
| 5 | 0.06 | 7 | 5 | 0.30 | | | | | |
| 6 | 0.03 | 7 | 6 | 0.20 | | | | | |
| 7 | 0.02 | 7 | 7 | 0.15 | | | | | |

input　　specified　　actual (output)

# Histogram Matching – Numerical example



| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | | | | |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | | | | |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | | | | |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | | | | |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | | | | |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | | | | |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | | | | |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | | | | |

input　　specified　　actual (output)

# Histogram Matching – Numerical example



- Step 3: find the corresponding value of $z_q$ so that $G(z_q)$ is closest to $s_k$

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | | | | |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | | | | |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | | | | |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | | | | |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | | | | |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | | | | |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | | | | |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | | | | |

input　　specified　　actual (output)

# Histogram Matching – Numerical example

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| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | 0→0,1,2 | | | |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | **1**→3 | | | |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | **2**→4 | | | |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | **3**→4 | | | |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | **4**→5 | | | |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | 5→5 | | | |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | 6→6 | | | |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | 7→7 | | | |

input specified actual (output)

# Histogram Matching – Numerical example



- Step 4: transform the equalized pixels $s_k$ into the corresponding values $z_q$

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | 0→0,1,2 | | | |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | **1**→3 | | | |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | **2**→4 | | | |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | **3**→4 | | | |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | **4**→5 | | | |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | 5→5 | | | |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | 6→6 | | | |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | 7→7 | | | |

input specified actual (output)

# Histogram Matching – Numerical example

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| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | 0→0,1,2 | 0→3 | | |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | **1**→3 | 1→4 | | |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | **2**→4 | 2→5 | | |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | **3**→4 | 3→6 | | |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | **4**→5 | 4→6 | | |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | 5→5 | 5→7 | | |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | 6→6 | 6→7 | | |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | 7→7 | 7→7 | | |

input　　specified　　actual (output)

# Histogram Matching – Numerical example



- Let's compute the histogram of the output image

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | 0→0,1,2 | 0→3 | 0 | |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | **1**→3 | 1→4 | 1 | |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | **2**→4 | 2→5 | 2 | |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | **3→4** | 3→6 | 3 | |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | **4→5** | 4→6 | 4 | |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | 5→5 | 5→7 | 5 | |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | 6→6 | 6→7 | 6 | |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | 7→7 | 7→7 | 7 | |

input specified actual (output)

# Histogram Matching – Numerical example



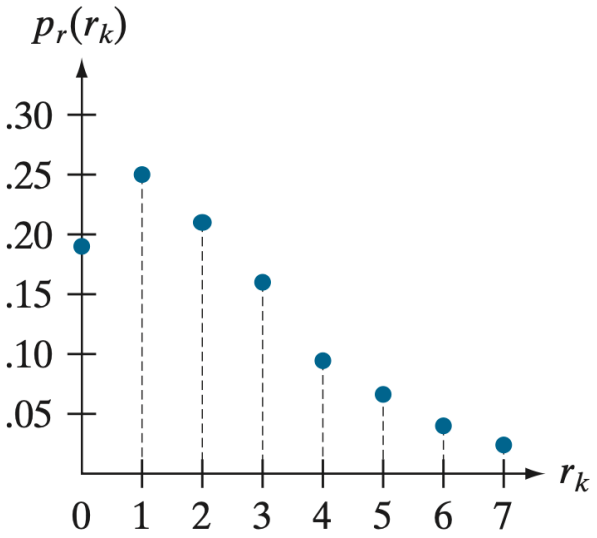
- We can compare it to the specified histogram

| $r_k$ | $p(r_k)$ | $s_k = T(r_k)$ | $z_k$ | $p(z_k)$ | $s_k = G(z_k)$ | | $s_k \rightarrow z_k$ | $r_k \rightarrow z_k$ | $z_k$ | $p(z_k)$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0.19 | 1 | 0 | 0 | 0.00 | 0 | 0→0,1,2 | 0→3 | 0 | 0 |
| 1 | 0.25 | 3 | 1 | 0 | 0.00 | 0 | **1**→3 | 1→4 | 1 | 0 |
| 2 | 0.21 | 5 | 2 | 0 | 0.00 | 0 | **2**→4 | 2→5 | 2 | 0 |
| 3 | 0.16 | 6 | 3 | 0.15 | 1.05 | 1 | **3→4** | 3→6 | 3 | 0.19 |
| 4 | 0.08 | 6 | 4 | 0.20 | 2.45 | 2 | **4→5** | 4→6 | 4 | 0.25 |
| 5 | 0.06 | 7 | 5 | 0.30 | 4.55 | 5 | 5→5 | 5→7 | 5 | 0.21 |
| 6 | 0.03 | 7 | 6 | 0.20 | 5.95 | 6 | 6→6 | 6→7 | 6 | 0.24 |
| 7 | 0.02 | 7 | 7 | 0.15 | 7.00 | 7 | 7→7 | 7→7 | 7 | 0.11 |

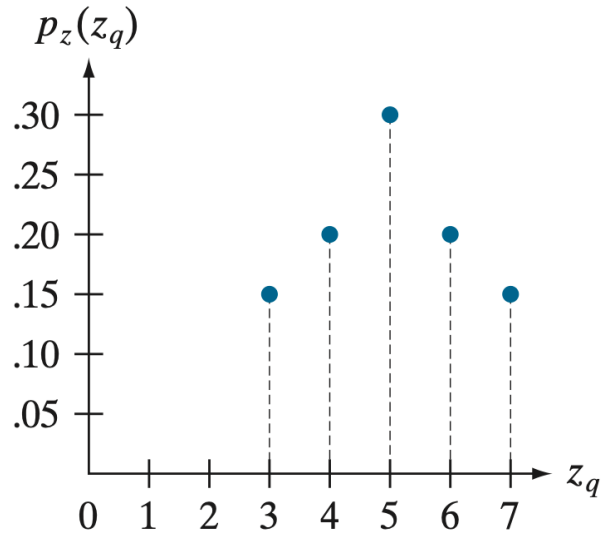
input　　specified　　actual (output)

# Histogram Matching – Numerical example

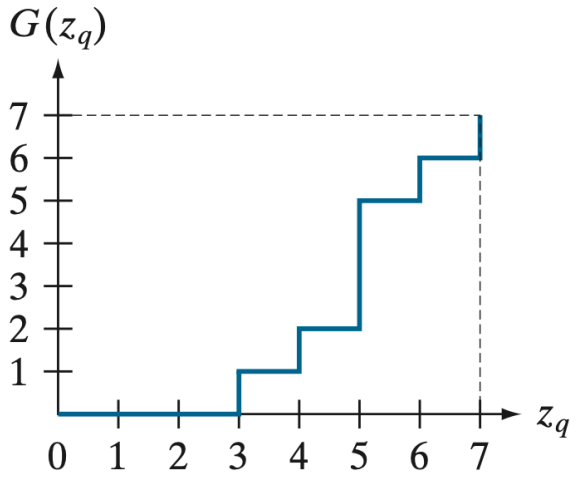




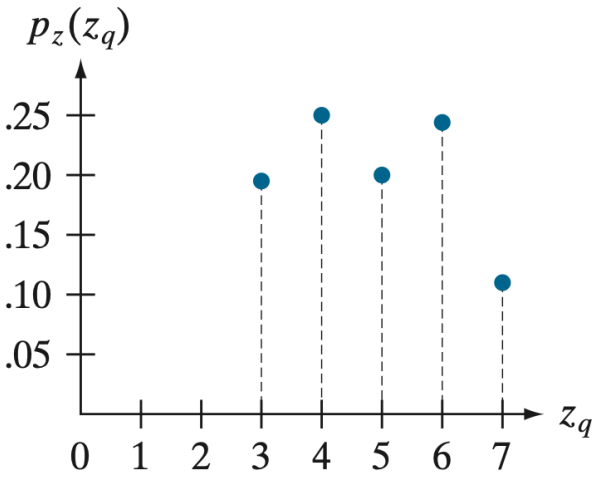

Histogram of input image




Specified histogram



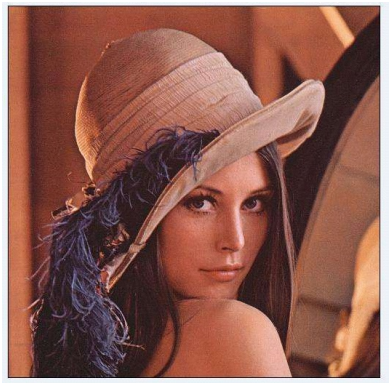

Transformation function



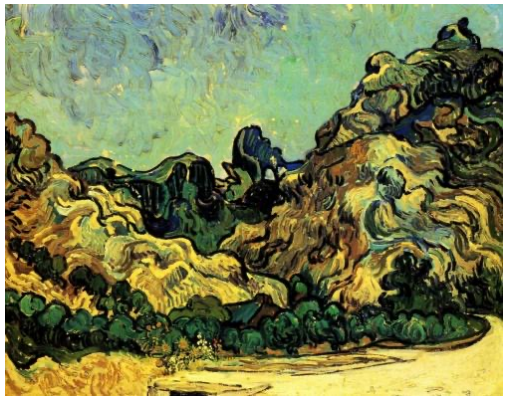

Histogram of output image

# Histogram Matching Application



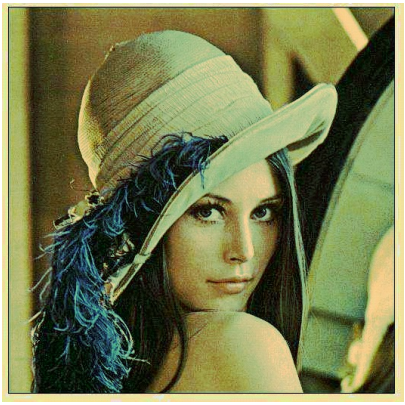


input





specified





output

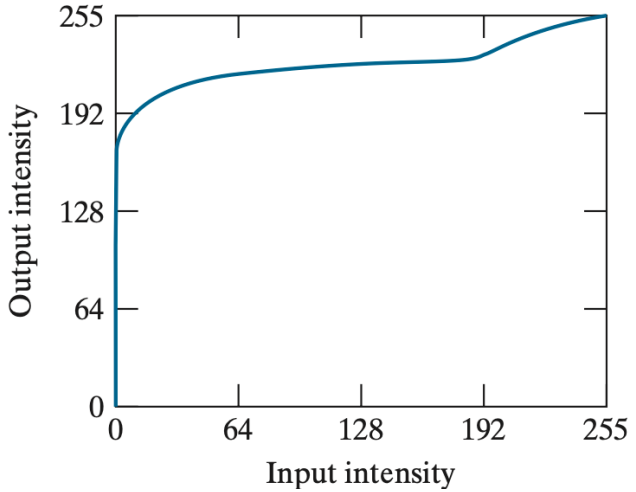
# Histogram Equalization VS Histogram Matching

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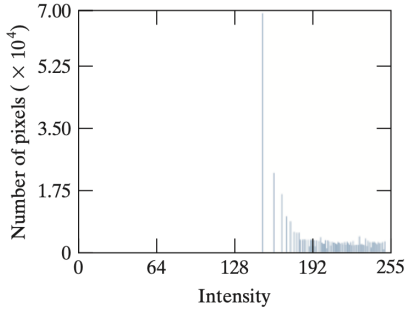
- Mars moon, Phobos, taken by NASA's Mars Global Surveyor



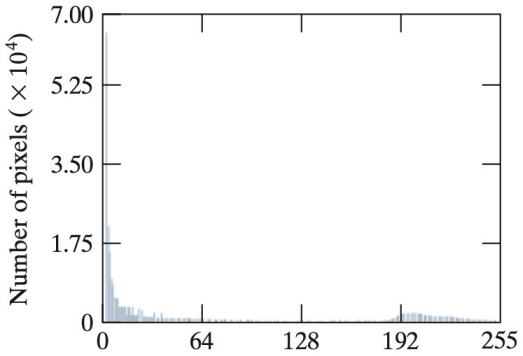
Histogram Equalization

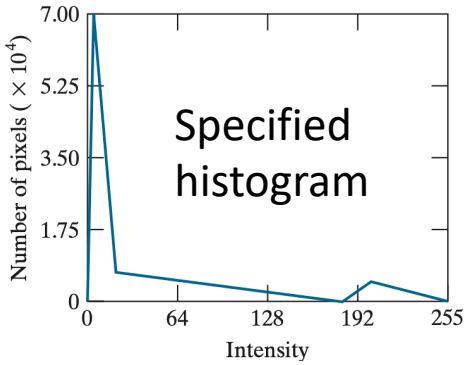


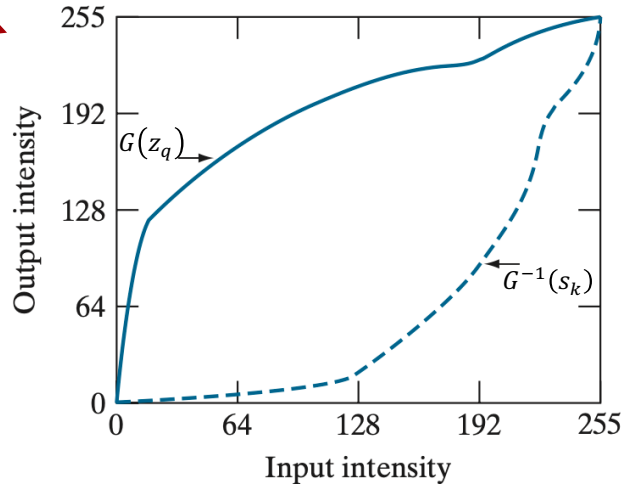


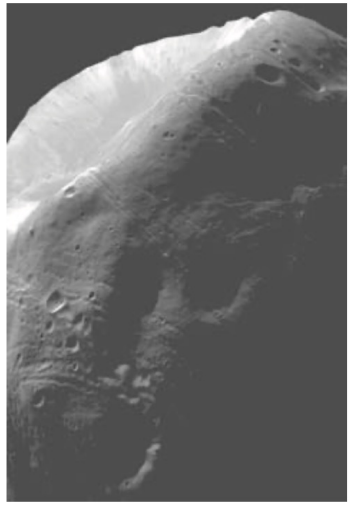



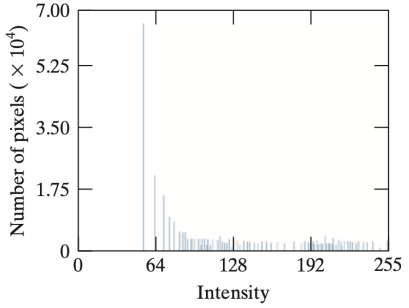

Histogram Matching





# Take home message



- Histogram describes intensity property of image, NO location (spatial) information.
- Simplest image processing technique: intensity transform.

$$s=T(r)$$

- The main purpose of intensity transform is to modify image histogram, to make the image contrast look more comfortable.
- Common intensity transform: contrast stretching, log, gamma, histogram equalization, histogram matching.

# Refining intensity transforms

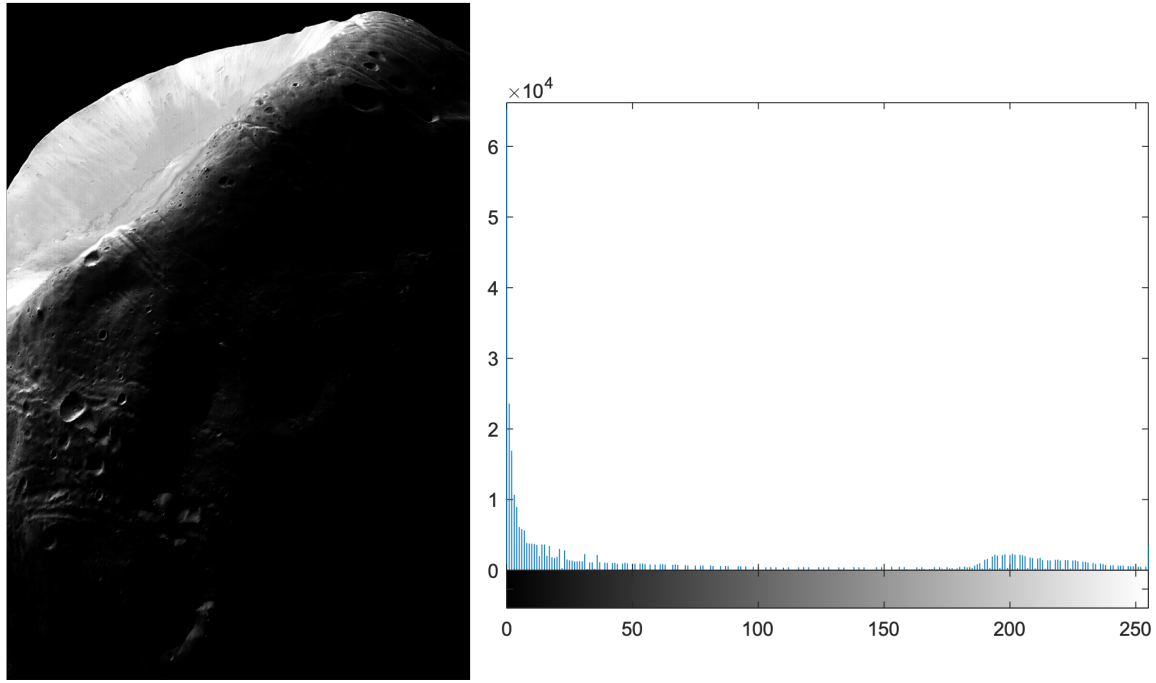


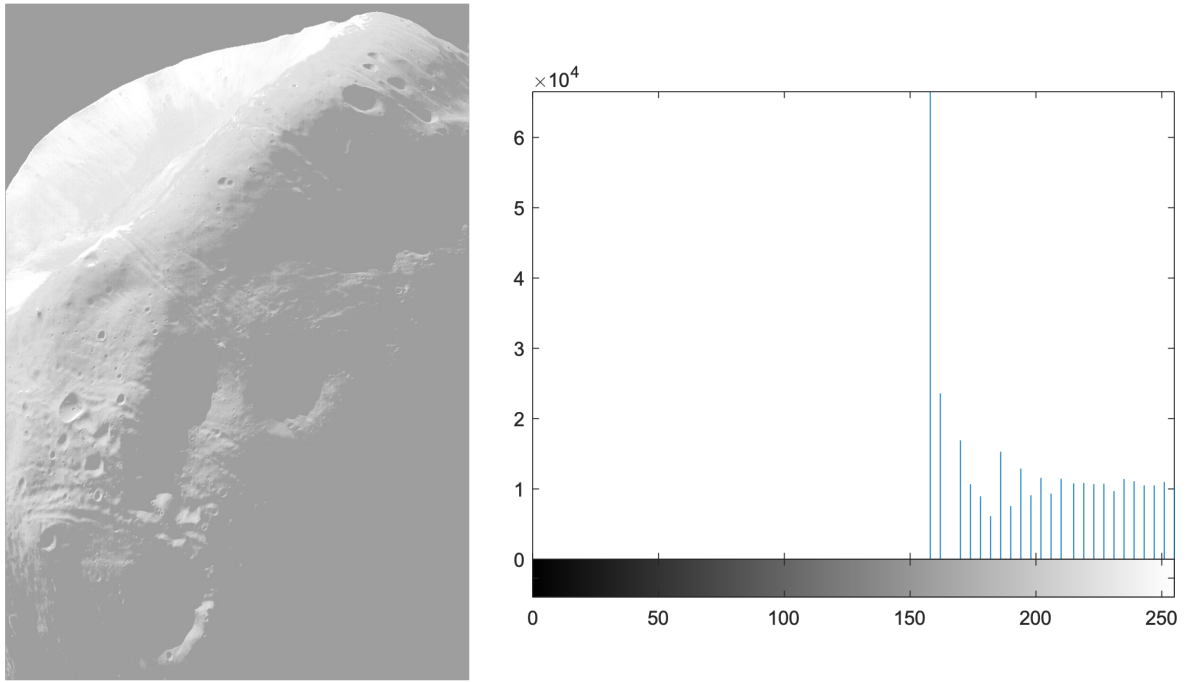
- Adaptive Histogram Equalization (AHE)
- Contrast Limited Adaptive Histogram Equalization (CLAHE)

# Key problem of Histogram Equalization (HE)



$$s = T(r) = (L-1)\int_0^r p_r(w)dw = (L-1)\sum_{j=0}^{k} p_r(r_j) = (L-1)\sum_{j=0}^{k}\frac{n_j}{MN} = \frac{L-1}{MN}\sum_{j=0}^{k} n_j$$

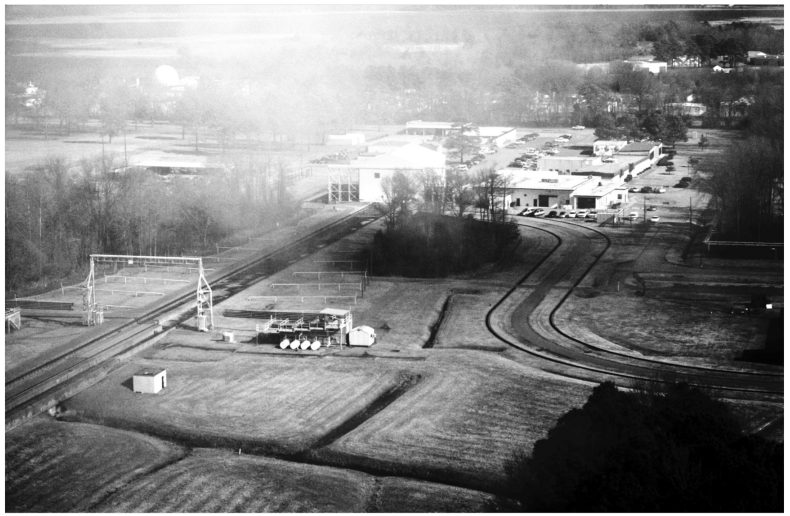


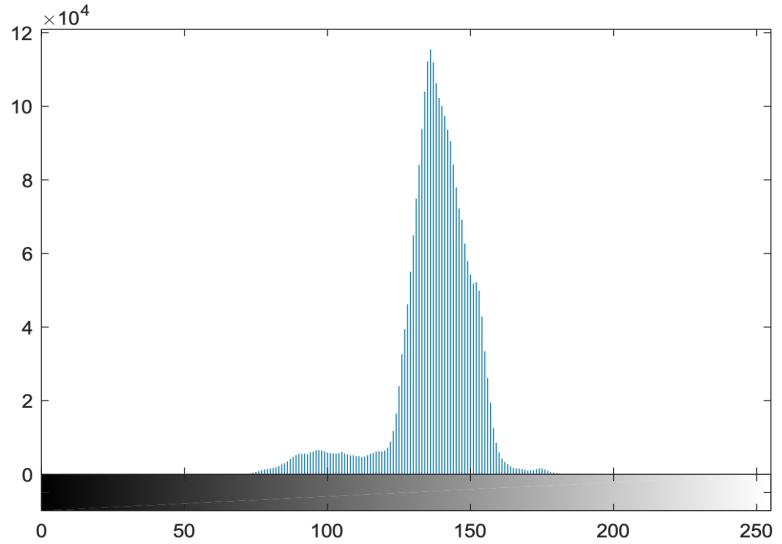



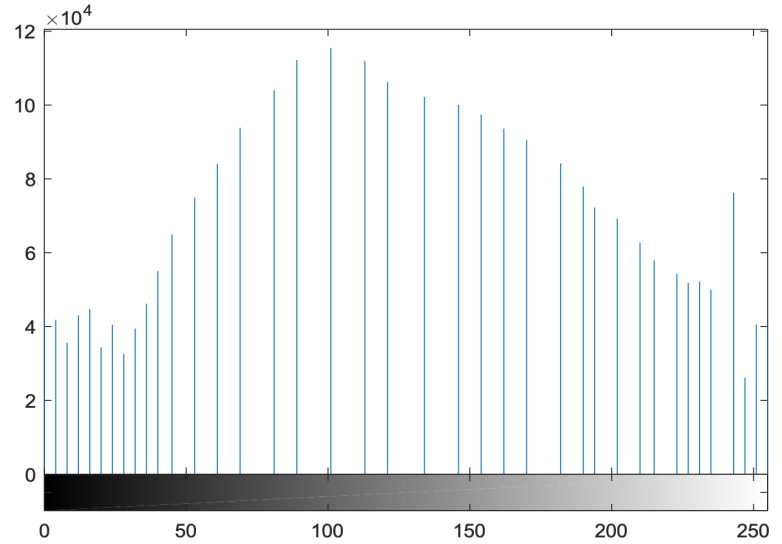
# Key problem of Histogram Equalization (HE)







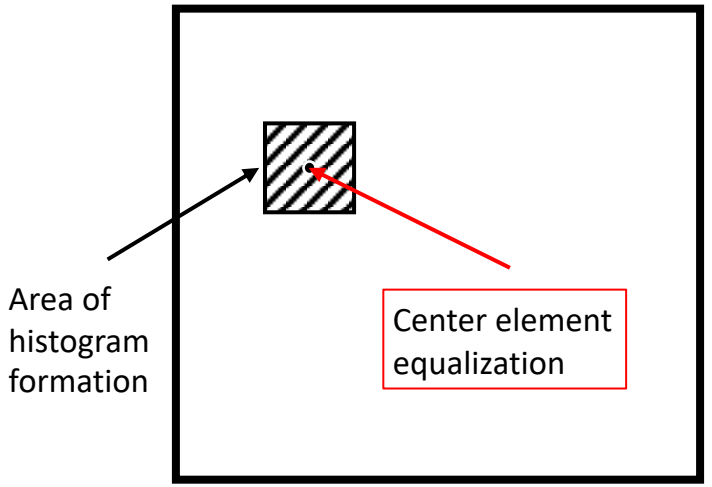


# Adaptive Histogram Equalization (AHE)



- Traverse every pixel with a $W * W$ patch, process histogram equalization within each patch and update the center pixel.
- Advantage: better uniform distributed histogram.
- Disadvantage: high complexity
- $O(W * W + L)$ within each patch
- $O(M * N * (W * W + L))$ for whole image

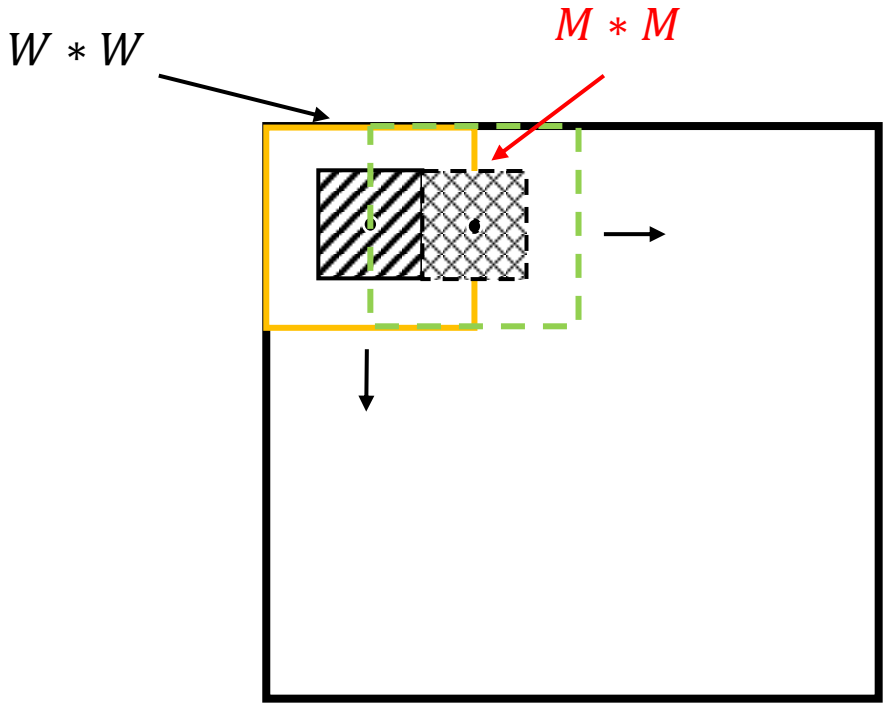


# Adaptive Histogram Equalization (AHE)



- For faster processing AHE, it is proposed to update a center patch of size $M * M$ instead of just the center pixel in each HE within the $W * W$ patch HE.

- Pixels near the image boundary have to be treated specially, This can be solved by extending the image by mirroring pixel lines and columns with respect to the image boundary.

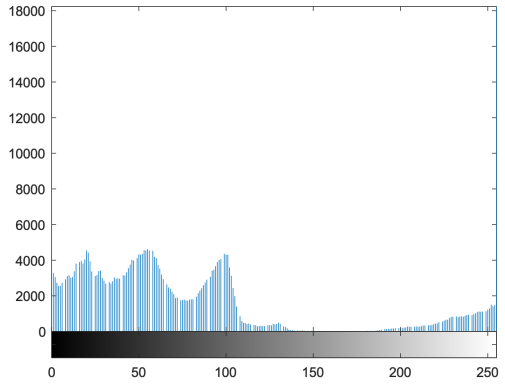


# Effect of AHE



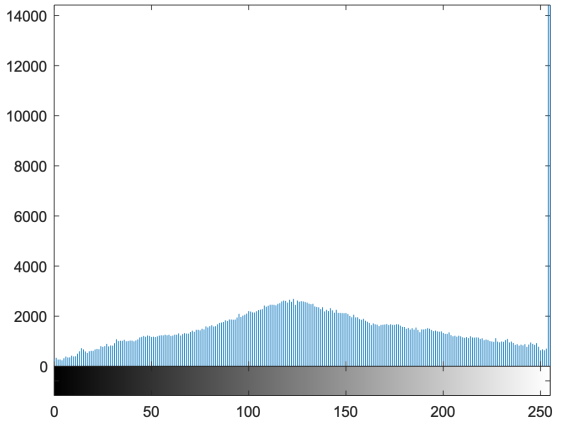
Raw





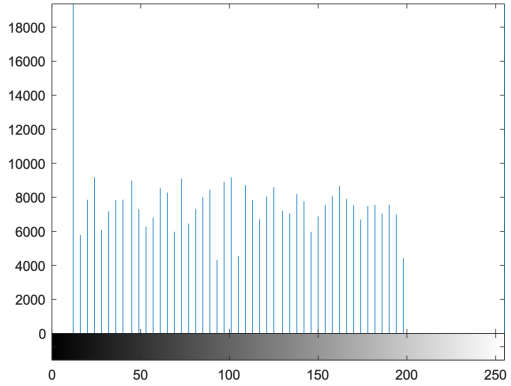

AHE






HE






- AHE has a tendency to overamplify noise in homogeneous areas
- Better at improving local contrast and edge information

# Contrast Limited Adaptive Histogram Equalization (CLAHE)

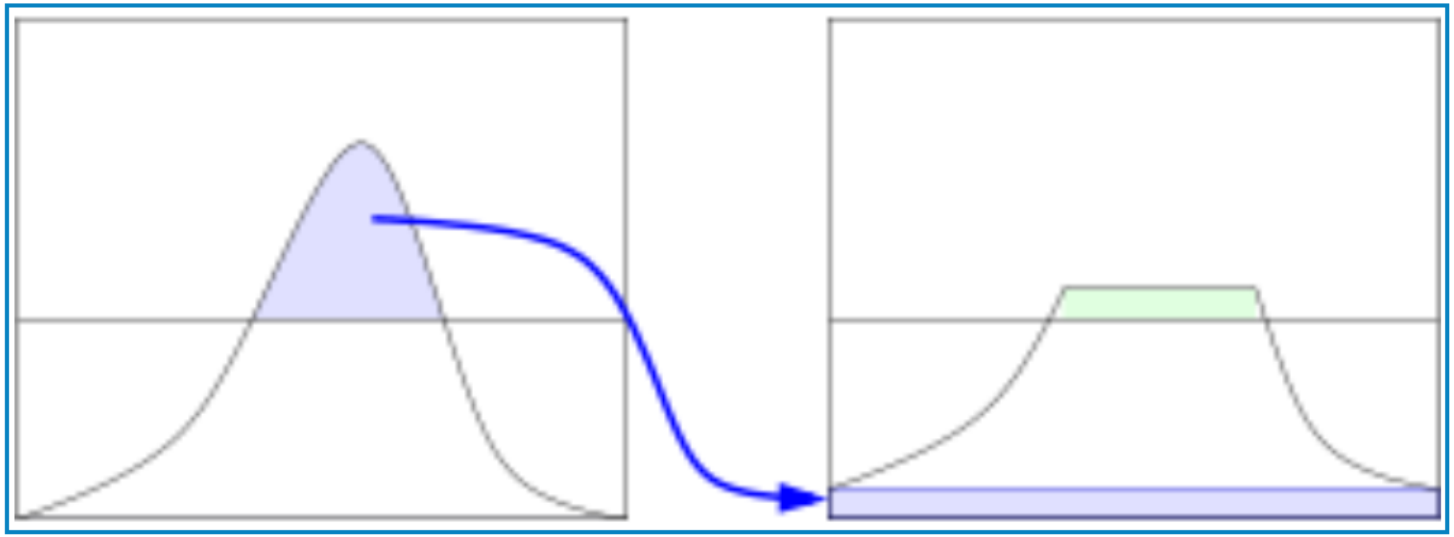


- CLAHE differs from naive AHE because it limits local contrast.
- CLAHE was developed to prevent the over amplification of noise that AHE can give rise to.
- This feature can also be applied to global histogram equalization, giving rise to contrast limited histogram equalization.

# Contrast Limited Adaptive Histogram Equalization (CLAHE)



- CLAHE limits the amplification by clipping the histogram at a predefined value before computing the CDF.
- This limits the slope of the CDF and therefore of the transformation function.
- The so-called clip limit depends on the normalization of the histogram and thereby on the size of the neighborhood region.

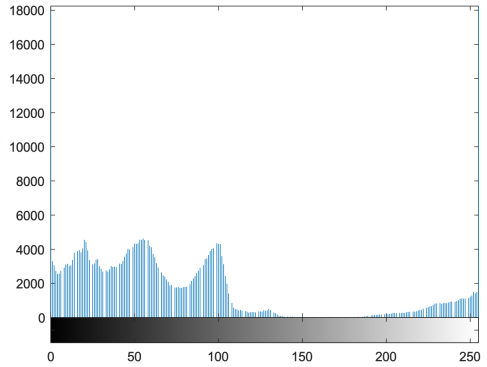


# CLAHE



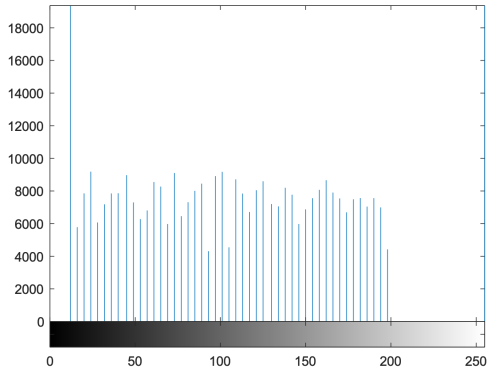
Raw





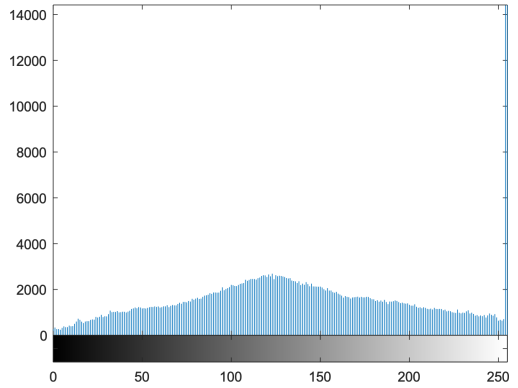

HE





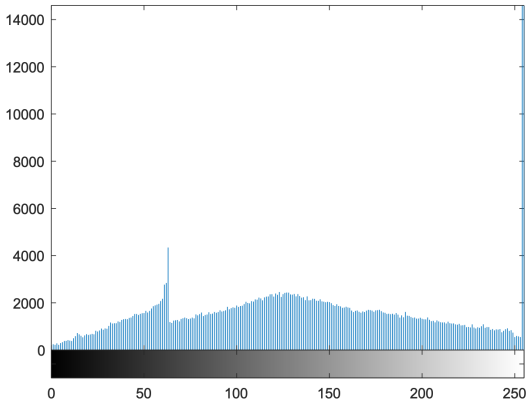

AHE






CLAHE





# Take home message



- Key idea: AHE&CLAHE were developed to prevent the over amplification of noise.
  - AHE: Fragments the bins to spread them
  - CLAHE: cuts-off the head of the histogram and distributes it

